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A PROJECT REPORT ENTITLED

DYNAMIC CONTEXT-AWARE TRIES FOR KNOWLEDGE GRAPH-CONSTRAINED LARGE LANGUAGE MODEL GENERATION

BY

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DECLARATION

We, BERNARD KIRK ADJANOR KATAMANSO, ERICA AMONOR, JOSEPH OSEI NYARKO and JESSICA AFUA ETORNAM NSAFOAH, declare that this project work is our own original work. It is being submitted for the degree of Bachelor of Science in Computer Science and Engineering at the University of Mines and Technology (UMaT), Tarkwa. It has not been submitted, either in whole or in part, for any degree or examination at any other university.

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ABSTRACT

Large language models (LLMs) are capable of multi-hop reasoning .....

DEDICATION

Dedicated to our parents, whose endless sacrifices, unwavering belief, and foundational support made this journey possible.

And to the machine, an elegant engine of probability, reminding us that predicting the word is not understanding the world.

ACKNOWLEDGEMENTS

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Finally, we acknowledge the broader academic community whose work has inspired and informed our research. This project stands on the shoulders of countless scholars, and we are grateful for their contributions to the field.

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LIST OF ABBREVIATIONS

DCA-Trie	Dynamic Context-Aware Trie
LLM	Large Language Model
KGQA	Knowledge Graph Question Answering
RAG	Retrieval-Augmented Generation
KG	Knowledge Graph
GCR	Graph-Constrained Reasoning
CWQ	ComplexWebQuestions
RLHF	Reinforcement Learning from Human Feedback
CoT	Chain-of-Thought
DoG	Decoding on Graphs
SIR	Semantic Irrelevance Ratio
GCD	Grammar-Constrained Decoding
QA	Question Answering
NLP	Natural Language Processing
AI	Artificial Intelligence
BFS	Breadth-First Search

CHAPTER 1

INTRODUCTION

1.1 Background

Large language models (LLMs) have emerged as one of the most consequential advances in artificial intelligence, demonstrating capabilities across language understanding, generation, and multistep reasoning that were considered beyond reach only a decade ago (Brown *et al.*, 2020; OpenAI *et al.*, 2024). Models such as the GPT series (Brown *et al.*, 2020; OpenAI *et al.*, 2024) and the Llama family (Touvron, Martin and Stone, 2023; Grattafiori *et al.*, 2024) achieve strong performance on a broad spectrum of language tasks by learning statistical regularities over vast corpora of text. The scale and breadth of this training enables these models to exhibit apparent knowledge across diverse domains, from history and science to legal reasoning and medical information, without any explicit representation of factual knowledge in their architecture.

Despite these achievements, a fundamental and persistent failure mode undermines the reliability of LLMs in knowledge-intensive settings: hallucination. Ji *et al.* (2023) define hallucination as the generation of content that is fluent and coherent but factually unsupported or contradicted by verifiable sources. Huang *et al.* (2025) provide a comprehensive taxonomy distinguishing intrinsic hallucination, where the output contradicts a provided source, from extrinsic hallucination, where the output cannot be verified against any source. Critically, hallucination is not a rare edge case. Luo *et al.* (2025a) report that state-of-the-art retrieval-augmented approaches produce hallucinated reasoning steps in approximately one-third of cases even when granted access to a supporting knowledge graph. This failure rate renders unconstrained LLMs unsuitable for high-stakes knowledge-intensive tasks without additional mechanisms to enforce factual grounding. The structural inevitability of hallucination has been formally established in recent theoretical work. Xu, Jain and Kankanhalli (2025) prove that for any computable ground truth function, no LLM training procedure can achieve zero hallucination probability across all inputs–hallucination is an inherent property of computable learning systems rather than a transient training artifact. This "inevitability" result underscores

that mitigation strategies relying solely on improved training, scaling, or prompting cannot fully eliminate hallucination; external structural constraints that ground generation in verifiable sources are theoretically necessary for guaranteed faithfulness. Knowledge graph-constrained decoding provides exactly such an external mechanism, making hallucination mathematically impossible rather than merely improbable.

The structural cause of hallucination is well understood. Autoregressive LLMs generate text by conditioning each token on all previously generated tokens, producing a probability distribution over the vocabulary at each step: $P(y_t \mid y_{<t}, x)$, where x is the input and $y_{<t}$ is the partially generated sequence (Brown *et al.*, 2020). This process is effective at producing locally coherent text but provides no mechanism for grounding each generated claim in a verifiable external source. An error introduced at step t propagates forward as the conditioning context for all subsequent tokens, producing confident and grammatically well-formed outputs whose factual incorrectness has no surface signal (Huang *et al.*, 2025; Ji *et al.*, 2023).

Knowledge graph question answering (KGQA) represents one of the most demanding contexts in which this failure mode manifests. A knowledge graph (KG) is a structured, human-curated database of verified facts encoded as (entity, relation, entity) triplets, for instance, (Marie Curie, award received, Nobel Prize in Physics) or (Paris, capital of, France), forming a large-scale graph over which multi-hop reasoning paths can be traced (Bollacker *et al.*, 2008). The Freebase knowledge graph, which provides the underlying resource for the standard KGQA benchmarks WebQSP (Yih *et al.*, 2016) and ComplexWebQuestions (Talmor and Berant, 2018), encodes tens of millions of such triplets covering a wide range of domains and relation types. Answering a multi-hop question such as 'What is the nationality of the director of the film in which the Blue Hawaii actor starred?' requires traversing a chain of three or more KG triplets in the correct order, a task for which an unconstrained LLM, relying entirely on parametric memory, may produce a plausible-sounding but factually incorrect chain that is indistinguishable in fluency from a correct one (Lan *et al.*, 2022a; He *et al.*, 2021a).

The dominant paradigm for addressing this problem is retrieval-augmented generation (RAG), in which relevant passages or subgraphs are retrieved from an external source and provided to the LLM as additional context (Lewis *et al.*, 2020; Izacard and Grave, 2021). While RAG reduces hallucination by providing relevant source material, it does not prevent the LLM from ignoring that material or generating claims that go beyond it. The LLM remains free to

produce any token at any step; the retrieved context is a soft influence on its output, not a hard constraint on what it may generate (Asai *et al.*, 2024; Agrawal *et al.*, 2024).

KG-constrained decoding addresses this limitation directly. Rather than providing the KG as a soft contextual influence, constrained decoding frameworks interpose a constraint oracle between the LLM's logit distribution and the token selection mechanism. At each generation step, the oracle computes a binary mask over the vocabulary, setting to zero the logit of any token that would not correspond to a valid continuation of a KG path. The masking formulation, introduced by Geng *et al.* (2023) in the context of grammar-constrained decoding, takes the form $\tilde{\alpha}_t = m_t \odot \alpha_t$, where α_t is the raw logit vector, $m_t \in \{0, 1\}^{|V|}$ is the validity mask, and $\odot$ denotes element-wise product. The generation process thus selects from a restricted set of valid tokens at every step, making structural hallucination, the generation of facts that do not exist in the KG, mathematically impossible rather than merely improbable (Willard and Louf, 2023; Geng *et al.*, 2023).

The leading implementation of this paradigm is Graph-Constrained Reasoning (GCR) (Luo *et al.*, 2025a), which achieves 92.6 Hits@1 on WebQSP and 75.8 Hits@1 on CWQ with verified 100% structural faithfulness. GCR constructs a KG-Trie, a prefix tree (Fredkin, 1960) over all KG paths within L hops of the question entities, obtained by breadth-first search, and uses it as the constraint oracle during beam-search decoding. Because every token in the generated reasoning chain must correspond to a valid prefix in the trie, the output is guaranteed to consist entirely of verified KG facts. This represents a qualitative improvement over both unconstrained LLMs and soft RAG approaches for the specific property of structural faithfulness.

However, a significant limitation remains. The KG-Trie in GCR is computed once, before generation begins, from the graph structure and the question entity set alone. It carries no information about the question semantics, the direction in which reasoning is proceeding, or which entities and relations have already been committed to in the partial generation $y_{<t}$. Formally, the valid token set W_{val} is a function $f(G, E_q)$, independent of q and $y_{<t}$, and remains identical across every generation step. For a question about a specific narrow fact, this trie may contain hundreds or thousands of structurally valid but semantically irrelevant paths, forcing the LLM to discriminate among them using its own parametric knowledge: precisely the hallucination-prone mechanism that constrained decoding was designed to suppress (Li

et al., 2025; Luo *et al.*, 2025a).

Decoding on Graphs (DoG) (Li *et al.*, 2025) introduces dynamic step-wise expansion, updating the constraint trie after each committed entity to include KG neighbors of the current node. This topological dynamism ensures the constraint set reflects the reasoning direction taken, addressing GCR's structural rigidity. DoG also employs question semantics, but crucially, only for *ranking* candidate paths during beam search–the valid token set itself remains determined solely by graph connectivity, admitting every neighbor regardless of relevance to the question or partial generation. The permissiveness problem therefore persists: thousands of structurally valid but semantically irrelevant paths remain in the valid set at each step, forcing the LLM to discriminate among them using parametric knowledge. Furthermore, DoG sacrifices the computational efficiency of pre-computed tries, rebuilding the constraint structure at each decoding step. These limitations motivate the DCA-Trie framework, which applies semantic filtering directly to the constraint oracle, ensuring that the valid token set at each step is both structurally faithful and semantically relevant.

1.2 Problem Statement

Existing KG-constrained decoding frameworks, including GCR (Luo *et al.*, 2025a) and DoG (Li *et al.*, 2025), construct their constraint oracles as functions $f(G, E_q)$ of the knowledge graph and question entities, evaluated prior to or independently of the sequential generation process. This design has been shown to guarantee structural faithfulness (every generated token corresponds to a verified KG triplet), but provides no mechanism for adapting the constraint to the semantic content of the question or the direction taken by the partial generation (Geng *et al.*, 2023; Willard and Louf, 2023).

The consequence is a fundamental conflation of two distinct requirements. Structural validity requires that every generated token correspond to a valid KG path prefix, a requirement all existing frameworks satisfy. Contextual relevance requires that the constraint set reflect the semantic focus of the reasoning at the current step, a requirement no existing framework satisfies. As a result, existing constraint oracles are simultaneously too broad, admitting structurally valid but semantically irrelevant paths that force the model to rely on parametric discrimination, and too rigid, failing to narrow in response to the evolving reasoning direction as generation progresses (Li *et al.*, 2025; Luo *et al.*, 2025a).

This project addresses the following specific problem: current KG-constrained decoding frameworks do not condition the valid token constraint on the input question semantics or the partial generation, leaving the LLM responsible for semantic discrimination among a large set of valid-but-irrelevant paths at every generation step. This residual reliance on parametric knowledge partially re-introduces the hallucination risk that constrained decoding was designed to eliminate, and is most pronounced in multi-hop settings where the number of valid paths grows exponentially with hop depth (Lan *et al.*, 2022a; He *et al.*, 2021a; Talmor and Berant, 2018).

Very recent work has independently addressed the efficiency-faithfulness trade-off in KGQA. RouterKGQA (Yuan *et al.*, 2026) proposes routing between specialized and general models with semantic filtering at the answer level, achieving strong performance with minimal inference cost. While RouterKGQA and DCA-Trie both employ semantic relevance scoring, they operate at fundamentally different architectural stages: RouterKGQA filters *answers* after generation, whereas DCA-Trie filters *tokens* during generation. This distinction is critical: DCA-Trie's constraint oracle ensures the LLM never sees semantically irrelevant tokens, reducing cognitive load and enabling deterministic faithfulness guarantees that post-hoc filtering cannot provide. Furthermore, DCA-Trie introduces the Semantic Irrelevance Ratio (SIR; Chapter 3), the first metric to quantify constraint permissiveness independently of downstream accuracy–an evaluation dimension not addressed by routing-based approaches.

To address these limitations, this project proposes DCA-Trie, upgrading the constraint oracle from $W_{\text{val}}(t) = f(G, E_q)$ to $W_{\text{val}}(t) = f(G, E_q, q, y_{<t})$. The upgraded framework must simultaneously satisfy three conditions: (i) structural faithfulness is preserved, every token admitted at each step corresponds to a valid KG path, as verified by exhaustive triplet checking against Freebase; (ii) semantic irrelevance is reduced, the proportion of the valid set consisting of paths irrelevant to the question decreases relative to GCR and DoG, as measured by the Semantic Irrelevance Ratio (SIR) introduced in this work; and (iii) answer recall is maintained: no gold answer path is erroneously excluded by the relevance filter, with false negative rate held below 5% on the WebQSP validation set.

1.3 Aim and Objectives

This project aims to design, implement, and evaluate a Dynamic Context-Aware Trie (DCA-Trie) framework for knowledge graph-constrained large language model generation, in which the valid token constraint at each decoding step is conditioned jointly on the knowledge graph structure, the input question semantics, and the partially generated reasoning chain, achieving improved answer accuracy on multi-hop KGQA benchmarks while maintaining 100% structural faithfulness to the underlying knowledge graph.

To achieve this aim, the following objectives have been set:

i. To characterize the permissiveness problem in existing KG-constrained decoding frameworks by defining and measuring the Semantic Irrelevance Ratio (SIR) of GCR's static KG-Tries on the WebQSP and CWQ benchmarks, stratified by hop depth.

ii. To design a semantic relevance scoring mechanism using pretrained sentence transformer embeddings (Reimers and Gurevych, 2019) that can filter candidate KG paths by joint relevance to the input question and partial generation at inference time, without task-specific fine-tuning.

iii. To implement a static semantic filtering variant (DCA-Trie v1) that applies the relevance scorer at KG-Trie construction time, reducing trie size before generation begins while maintaining a false negative rate below 5% on gold answer paths on the WebQSP validation set.

iv. To implement a step-wise dynamic expansion variant (DCA-Trie v2) in which the trie is updated after each committed triplet using the relevance scorer conditioned jointly on the question and the partial generation $y_{<t}$, directly operationalizing the $W_{\text{val}}(t) = f(G, E_q, q, y_{<t})$ formulation.

v. To evaluate DCA-Trie v1 and DCA-Trie v2 against the GCR baseline and an unconstrained chain-of-thought (CoT) baseline on WebQSP and CWQ, measuring Hits@1, F1, structural faithfulness, SIR, and average trie size, with results stratified by question hop depth.

vi. To demonstrate the practical applicability of DCA-Trie by building an experimental

interactive prototype system (web-based or command-line QA interface) for research demonstration, allowing users to input complex natural language questions and receive factually grounded reasoning chains and answers, thereby showcasing the framework's real-world value beyond benchmark metrics.

1.4 Tools and Facilities Used

1.4.1 Facilities

The following facilities were utilized in the execution of this project:

i. Google Colab Pro (A100, 40 GB VRAM) for all model inference and constrained decoding experiments.

ii. University library and online academic databases (ACM Digital Library, arXiv, Google Scholar, ACL Anthology) for literature review and citation verification.

iii. University internet infrastructure for cloud connectivity, dataset access, and retrieval of model checkpoints from the HuggingFace model hub.

iv. Shared team repository hosted on GitHub for version control, experiment logging, and reproducibility of all reported results.

1.4.2 Software Tools

Table 1.1 presents an evaluation of the software tools implemented in this project, their role within the DCA-Trie framework, and the rationale for their selection.

Table 1.1 Implemented Software Tools Evaluation.

Tool	Use in Project	Rationale
Python / HuggingFace Transformers	Model inference pipeline; loading and running GCR's Llama-3.1-8B checkpoint; integrating the relevance scorer	Provides reliable model loading, tokenization, and generation utilities, and enables direct use of the released GCR checkpoint.
GCR Framework	Primary baseline; KG-Trie construction, graph-constrained decoding, and result reproduction on WebQSP and CWQ	Provides a strong baseline and supports the trie-based constrained decoding mechanism that DCA-Trie extends.
Sentence-transformers / all-MiniLM-L6-v2	Semantic relevance scoring; computing cosine similarity between question-context and candidate KG path embeddings	Provides fast semantic embeddings without extra fine-tuning and supports efficient iterative decoding.
PyTorch	Deep learning framework underpinning all model inference and custom decoding logic	Supports implementation and testing of custom decoding logic and integrates smoothly with HuggingFace.
Freebase / WebQSP and CWQ Datasets	Knowledge graph source and KGQA benchmarks for evaluation	Provides standard datasets for this research area and enables direct comparison with prior work.
Google Colab Pro (A100, 40 GB)	GPU environment for all inference experiments; Llama-3.1-8B constrained decoding with beam search	Provides enough GPU memory for stable experiments and removes the need for local high-end hardware.
Visual Studio Code + Git / GitHub	Development environment and version control for all four team members	Supports day-to-day coding, team collaboration, version tracking, and reproducibility throughout the project.

1.5 Methods Used

The project employs the following methods:

i. Systematic review and critical analysis of the academic literature on constrained decoding, knowledge graph question answering, hallucination in LLMs, and retrieval-augmented generation.

ii. Formal gap analysis of the four foundational papers, GCR (Luo *et al.*, 2025a), DoG (Li *et al.*, 2025), GCD (Geng *et al.*, 2023), and Outlines (Willard and Louf, 2023), to identify the specific limitations that motivate the DCA-Trie design, including the permissiveness problem and the absence of generation-state feedback in the constraint oracle.

iii. Formal specification of the DCA-Trie framework, including definition of the Semantic Irrelevance Ratio metric, the relevance scoring mechanism using sentence transformer embeddings (Reimers and Gurevych, 2019), the static filtering procedure for v1, and the step-wise dynamic expansion rule for v2.

iv. Reproduction of the GCR baseline on the WebQSP benchmark using GCR's publicly released Llama-3.1-8B checkpoint to establish a verified starting point within 1–2% of published Hits@1.

v. Implementation of DCA-Trie v1 and v2 as extensions to GCR's inference pipeline, with relevance threshold selected on a 100-question held-out validation subset of WebQSP to minimize false negative rate on gold answer paths.

vi. Quantitative evaluation of all system configurations on WebQSP and CWQ using Hits@1, F1, faithfulness rate, SIR, and average trie size per step, stratified by question hop depth following the evaluation protocol of Lan *et al.* (2022a).

vii. Ablation studies examining the sensitivity of DCA-Trie's performance to the relevance threshold τ, the sentence encoder architecture, and beam size k, isolating the contribution of each mechanism to the overall result.

viii. Qualitative error analysis identifying failure cases in which DCA-Trie's semantic filter

erroneously excludes a gold answer path, supporting honest characterization of the framework's limitations and providing concrete directions for future work.

1.6 Scope of Work

This project designs and evaluates a Dynamic Context-Aware Trie framework for KG-constrained LLM decoding, with the specific aim of addressing the permissiveness problem identified in GCR and DoG. The framework is developed and evaluated in the context of multi-hop KGQA over Freebase-grounded benchmarks (WebQSP and CWQ), which are the standard evaluation settings in the KG-constrained decoding literature. All experiments use GCR's publicly released Llama-3.1-8B checkpoint; no new model training or fine-tuning is performed. The semantic relevance scorer uses the all-MiniLM-L6-v2 sentence transformer without domain-specific adaptation. We emphasize that all experiments use GCR's publicly released Llama-3.1-8B checkpoint (Luo *et al.*, 2025a), which includes their fine-tuned KG-specialized parameters. No additional model training or fine-tuning is performed in this work; our contribution is strictly the constraint oracle architecture and semantic filtering mechanism.

The scope is explicitly bounded as follows. Generalization of DCA-Trie to knowledge graphs other than Freebase, or to non-KGQA tasks such as entity linking or structured information extraction, is identified as future work and is not evaluated in this dissertation. Formal proofs of the faithfulness guarantee under dynamic pruning are beyond scope; empirical measurement of the gold answer false negative rate is provided as evidence. The confidence-conditioned constraint tightening mechanism, in which constraint granularity is modulated by the LLM's token-level entropy, is described as a theoretical extension but is not implemented. Production-scale deployment, latency optimization for online serving, and integration with proprietary enterprise knowledge graphs are engineering extensions beyond the research scope of this project.

The primary contribution of this work is the DCA-Trie framework and the Semantic Irrelevance Ratio metric. Secondary contributions are the empirical characterization of constraint permissiveness in existing frameworks and the ablation analysis isolating the contribution of semantic filtering from dynamic expansion. All code, experimental configurations, and the

curated evaluation datasets will be made publicly available to support reproducibility and future extension.

1.7 Organization of Work

Chapter 1 introduces the study by identifying the core problem, stating the project objectives, and defining the intended DCA-Trie solution, thereby establishing the motivation, scope, and overall direction of the research. Chapter 2 presents the literature review by examining prior work on hallucination in LLMs, KGQA, constrained decoding, and related frameworks, and it identifies key limitations in existing approaches to justify the need for a context-aware constraint mechanism. Chapter 3 combines problem formulation and methodology, defining the upgraded constraint setup, research questions, evaluation criteria, and implementation strategy for DCA-Trie v1 and v2 within the GCR pipeline (Luo *et al.*, 2025a). Chapter 4 presents results and discussion, reporting quantitative evaluation across WebQSP and CWQ, analysing faithfulness and semantic irrelevance trends, and interpreting the findings through ablation and error analysis. Chapter 5 concludes the dissertation by summarizing the main contributions, highlighting practical implications, and presenting recommendations for future work.

CHAPTER 2

LITERATURE REVIEW

While large language models (LLMs) excel at many tasks, their reliance on autoregressive generation makes them prone to hallucinating fluent but false information (Ji *et al.*, 2023). This unreliability is a major roadblock for knowledge-intensive question answering, where verifiable reasoning is essential. To address this, recent research has focused on grounding models in structured external facts via knowledge graphs. Even so, forcing a model to stay strictly faithful to these graphs during generation, without breaking its natural fluency, remains a complex challenge.

This literature review examines existing research on these challenges and explores the potential role of knowledge graph constraints in addressing them and ensuring factual reliability during multi-hop reasoning. Rather than describing the proposed solution immediately, it maps the intersection of language model hallucination, multi-step validation, and structured grounding. By evaluating current mitigation strategies, from advanced prompting to static decoding, this chapter highlights where soft-grounding approaches fall short, establishing the specific technical gaps this project addresses.

2.1 Large Language Models

To understand how knowledge graph constraints can be applied to language models, it is first necessary to understand the underlying mechanism of language model generation. Large language models (LLMs) are the foundational engines driving modern natural language processing. This section establishes the architectural and generative properties of LLMs, specifically detailing the autoregressive process that gives rise to both their fluent reasoning capabilities and their vulnerability to factual hallucination.

2.1.1 Transformer Architecture

Modern LLMs are built upon the Transformer architecture, introduced by Vaswani *et al.* (2017). At its core, the Transformer relies on a scaled dot-product self-attention mechanism that allows the model to dynamically weigh the importance of different tokens in an input

sequence regardless of their positional distance. Formally, for a set of queries Q, keys K, and values V, the attention output is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2.1}$$

where d_k represents the dimensionality of the keys. This represented a fundamental shift from prior recurrent architectures, enabling highly parallelized training and the capture of long-range dependencies. While the original framework proposed an encoder-decoder structure designed for translation tasks, the field has largely converged on decoder-only variants for text generation. In a decoder-only model, self-attention is causally masked, meaning that when computing the representation for a given token, the model can only attend to previous tokens and not future ones. This strictly left-to-right processing naturally aligns with the task of predicting the next word in a sequence. The Llama-3.1-8B model (Dubey *et al.*, 2024) is a prominent example of this decoder-only, causally-masked transformer design, and has been adopted as the backbone in recent KG-constrained decoding work (Luo *et al.*, 2025a).

2.1.2 Autoregressive Generation

The text production process in these models is fundamentally autoregressive. Given a vocabulary $\mathcal{V}$, an input sequence x, and previously generated output tokens $y_{<t}$, the model computes a probability distribution over the entire vocabulary for the next token: $P(y_t|y_{<t}, x)$. Because the generation condition includes all previously generated tokens, every new word becomes part of the context for the next word. Decoding strategies, such as beam search and nucleus sampling (Holtzman *et al.*, 2020), determine how a specific token is selected from this computed distribution. Crucially, because this distribution is always computed over the full vocabulary, any mathematically possible token has a strictly positive, non-zero probability of being generated at any step. This autoregressive property is the mechanism that produces highly fluent, coherent text; however, because each step conditions on prior outputs without an external verifier, it is also the direct structural cause of hallucination.

2.1.3 Large-Scale Pretraining

The transition from basic transformers to "large" language models is driven by scale in both parameter count and training data. Foundational models are pretrained on massive corpora of unlabelled text using a simple self-supervised objective, typically predicting the next token. The GPT series (Brown *et al.* 2020; OpenAI *et al.* 2024) and the open-weight Llama series (Touvron, Martin and Stone 2023; Dubey *et al.* 2024) demonstrated that scaling this simple objective leads to the emergence of powerful downstream capabilities, including few-shot learning and broad world knowledge. During pretraining, the model internalizes patterns, facts, and logical structures present in its training distribution. However, this parametric memory is fundamentally static and statistical. While scaling up models improves their fluency, syntax, and ability to recall commonly observed facts, it does not fix the underlying generation mechanism. The model still lacks a structural way to interact with dynamic external facts or guarantee that its statistical outputs map to verified truth in the real world.

2.1.4 Instruction Following and Chain-of-Thought

To align these pretrained base models with user intent, they undergo post-training processes such as instruction tuning and reinforcement learning from human feedback (RLHF). This alignment allows models to engage in dialogue and follow complex instructions. One of the most significant capabilities elicited from aligned LLMs is multi-step reasoning, typically prompted via Chain-of-Thought (CoT; Bosma *et al.* 2022). As surveyed by Huang and Chang (2023), CoT prompting encourages the model to generate intermediate reasoning steps before arriving at a final answer, effectively breaking down complex problems. While CoT dramatically improves performance on logic and reasoning benchmarks, it suffers from the same autoregressive limitation: each intermediate reasoning step is generated using only the model's internal statistical distributions. Therefore, while CoT helps structure a language model's reasoning process, it cannot logically ground those outputs in verifiable sources. This persistent lack of factual grounding creates the necessity for more rigorous interventions, such as knowledge graph-constrained decoding.

2.2 Hallucination in Large Language Models

2.2.1 Definition and Taxonomy

Hallucination in natural language generation refers to the production of content that is fluent and internally coherent yet factually unsupported by or in outright contradiction with verifiable sources (Ji *et al.*, 2023). Huang *et al.* (2023) refine this definition into a two-part taxonomy. *Intrinsic hallucination* occurs when the model output directly contradicts material present in the provided source context. *Extrinsic hallucination* occurs when the output contains claims that cannot be verified against any external source, even if not directly contradicting a specific reference.

Both types are documented at non-trivial rates in deployed systems. Ji *et al.* (2023) survey over 30 works across summarization, dialogue, and question answering, finding hallucination rates ranging from 15% to over 60% depending on task type and measurement methodology. In the specific context of knowledge graph question answering, Luo *et al.* (2025a) report that retrieval-augmented reasoning approaches produce hallucinated reasoning steps in approximately one-third of cases even when provided direct access to a supporting knowledge graph, underscoring that the problem is not simply one of insufficient external knowledge.

2.2.2 Structural Causes: The Autoregressive Generation Mechanism

The root cause of hallucination lies in the autoregressive generation mechanism shared by all modern large language models. At each generation step, the model computes a probability distribution over its vocabulary conditioned on the input and all previously generated tokens. Three structural properties of this mechanism make hallucination an ever-present risk rather than an occasional failure.

First, the generation distribution is computed entirely from learned parameters, with no term that corresponds to a factual verification operation against an external source. The model has no mechanism to "check" a candidate token against a knowledge base before selecting it. Second, errors propagate forward: because each step is conditioned on all prior output, a hallucinated token at step t becomes part of the conditioning context for step $t+1$, producing fluent and internally consistent continuations of an already-incorrect chain. Third, the distribution is defined over the full vocabulary at every step, meaning that any token has a

strictly positive probability of selection at any step, regardless of factual correctness (Brown *et al.*, 2020; Huang *et al.*, 2023; Ji *et al.*, 2023).

2.2.3 The Scaling Paradox

A common objection to the severity of the hallucination problem is that sufficiently large models should eventually achieve adequate factual reliability. The evidence does not support this view. Lin, Hilton and Evans (2022) introduce TruthfulQA, a benchmark targeting questions that frequently elicit plausible but false responses. Their central finding is that larger models do not outperform smaller models on truthfulness and in certain categories exhibit *inverse scaling*; in other words, larger models perform worse because they more effectively reproduce prevalent but incorrect beliefs from training data.

Perez *et al.* (2023) identify a compounding problem in models trained with reinforcement learning from human feedback: such models learn to generate confident-sounding responses that align with apparent user expectations rather than with verifiable facts, decoupling confidence from accuracy. Kandpal *et al.* (2023) show that while larger models memorize more training-data facts and improve on frequently-seen entity combinations, they do not improve on compositional questions requiring combinations of facts not seen during training. This precisely mirrors the structure of multi-hop knowledge graph questions. Mallen *et al.* (2023) confirm substantial hallucination rates for GPT-4 on questions involving tail entities, regardless of model scale. Together, these findings establish that hallucination is a structural feature of the autoregressive mechanism, not a scale problem with a straightforward remedy.

2.3 Multi-Step Reasoning: Chain-of-Thought and Its Limits

2.3.1 Chain-of-Thought Prompting

Chain-of-thought (CoT) prompting, introduced by Bosma *et al.* (2022), is the dominant paradigm for eliciting multi-step reasoning from large language models. By including worked examples in the prompt that show intermediate reasoning steps, the model learns to decompose complex problems into a sequence of simpler sub-problems before producing a final answer. Bosma *et al.* (2022) demonstrate substantial accuracy improvements on arithmetic, commonsense, and multi-hop question answering benchmarks, with the capability emerging at approximately 100 billion parameters. Zero-shot chain-of-thought (Kojima

et al., 2022) achieves similar benefits without worked examples by adding the phrase "Let's think step by step" to the prompt, demonstrating that the underlying capability is latent in sufficiently large models and requires only elicitation.

The limitation of CoT is that each intermediate step is generated by the same autoregressive mechanism described in the previous section. An incorrect intermediate step propagates forward as conditioning context, and no mechanism verifies that any step corresponds to a fact in an external knowledge source. A CoT chain can be entirely fluent and internally coherent while containing one or more steps that are factually fabricated.

2.3.2 Extensions: Self-Consistency and Tree-of-Thought

Two major extensions of chain-of-thought address its reliability limitations without resolving its structural faithfulness problem. Self-consistency (Wang *et al.*, 2023c) samples multiple independent CoT chains and selects the final answer by majority vote, marginalizing over individual chain errors. This approach substantially improves accuracy on benchmarks such as GSM8K and multi-hop QA by reducing sensitivity to individual hallucinated chains. However, it provides no guarantee that any single chain is factually grounded: a majority of independently hallucinated chains can still produce an incorrect majority-vote answer, particularly on deep multi-hop questions where error probabilities compound.

Tree-of-Thought (Cao *et al.*, 2023a) frames generation as a search over a tree of partial reasoning sequences, using the language model as both a generator and an evaluator to prune unpromising branches. This improves performance on tasks requiring backtracking and exploration. The fundamental limitation remains: pruning decisions are made by the model itself, using its own parametric knowledge as the sole evaluator, without any external verification of the factual status of candidate reasoning steps.

2.3.3 The Faithfulness Ceiling of Prompting-Based Approaches

The common limitation of all prompting-based approaches, including few-shot, chain-of-thought, self-consistency, tree-of-thought, and their combinations, is that they modify the *input* to the language model without modifying the *generation mechanism*. Because decoding still occurs over the full vocabulary, every token retains a strictly positive probability at every step. True structural faithfulness, defined as the guarantee that every generated token

corresponds to a verified external fact, requires that invalid tokens be assigned exactly zero probability. This can only be achieved by intervening directly in the decoding algorithm, which is the function of constrained decoding approaches (Geng *et al.*, 2023; Willard and Louf, 2023).

2.4 Knowledge Graphs and Knowledge Graph Question Answering

2.4.1 Knowledge Graphs

A knowledge graph is a structured database of verified facts, represented as a set of directed, labelled triplets (e, r, e') where $e, e' \in \mathcal{E}$ are entities and $r \in \mathcal{R}$ is a relation type. Each triplet encodes a single human-verified fact: for example, (*Marie Curie*, *award_received*, *Nobel_Prize_Physics*) or (*Blue_Hawaii*, *film.film.featured_film_location*, *Hawaii*). Figure 2.1 illustrates a simplified fragment of such a structure, demonstrating how these triplets interlink to form a broader network. Freebase (Bollacker *et al.*, 2008), the knowledge graph underlying the evaluation benchmarks used in this work, encodes tens of millions of such triplets across geography, history, entertainment, and science, and has been widely adopted as the standard resource for KGQA research (Yih *et al.*, 2016; Talmor and Berant, 2018).

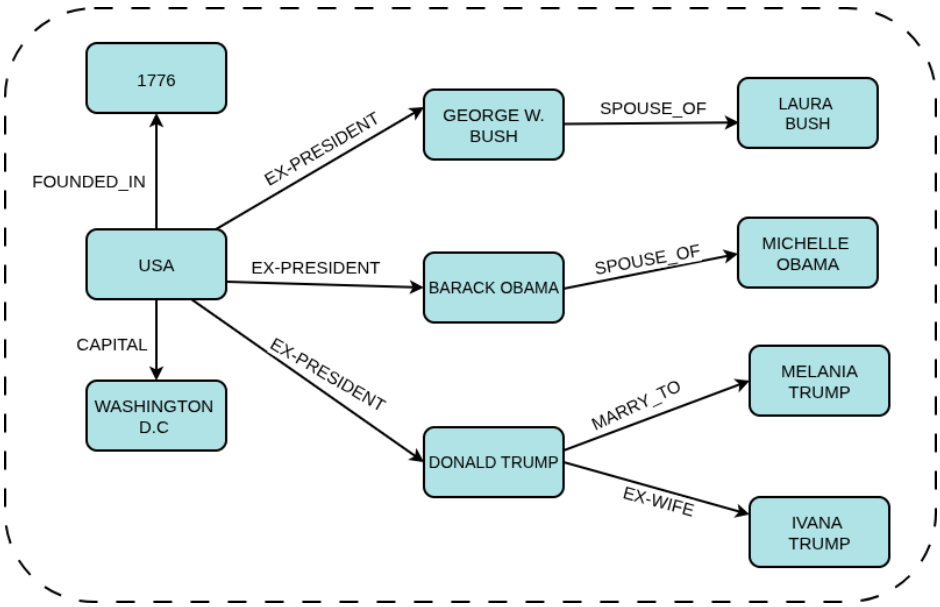

Figure 2.1 A simplified example of a knowledge graph fragment. Each triplet (edge) encodes a verified fact about the connected entities (nodes).

The critical distinction between a knowledge graph and an unstructured text corpus is that every triplet in the KG has been explicitly verified and is machine-readable in a deterministic,

unambiguous form. This property makes KGs suitable as sources for hard constraints on generation: a candidate token can be verified against the KG via an exact lookup, rather than via probabilistic semantic matching against raw text. A multi-hop reasoning path of length L through the KG is a sequence $(e_0, r_1, e_1, \ldots, r_L, e_L)$ in which each consecutive triplet (e_{i-1}, r_i, e_i) is a verified member of $\mathcal{T}$ (Lan *et al.*, 2022b).

2.4.2 Knowledge Graph Question Answering

Knowledge graph question answering (KGQA) requires a system to identify the answer entity e_L to a natural language question q by reasoning through a chain of verified KG triplets. The task is formally defined as: given question q with associated question entities $\mathcal{E}_q \subseteq \mathcal{E}$ identified by entity linking, identify the terminal entity of the correct reasoning path in $\mathcal{G}$ (Lan *et al.*, 2022b).

Early KGQA approaches relied on either semantic parsing (converting questions to structured SPARQL queries) or information retrieval over KG subgraphs (Lan *et al.*, 2022b). More recent approaches use large language models as retrievers, agents iteratively querying the KG, or generators conditioned on retrieved subgraph context (Jiang *et al.*, 2023a; Jin *et al.*, 2024; Luo *et al.*, 2024). The shift toward LLM-based KGQA has improved fluency and coverage but reintroduced the hallucination risk that structured approaches avoided: LLM-generated reasoning chains are not guaranteed to correspond to verified KG paths.

2.4.3 Evaluation Benchmarks

Two benchmark datasets provide the evaluation setting for KGQA research. WebQSP (Yih *et al.*, 2016) is derived from the WebQuestions dataset and comprises 4,737 questions grounded in Freebase with reasoning depths of one to two hops, annotated with SPARQL queries and evaluated using Hits@1 and F1. ComplexWebQuestions (CWQ; Talmor and Berant 2018) extends WebQSP by compositionally transforming questions through conjunction, composition, and comparative operations, yielding approximately 34,000 questions requiring up to four reasoning hops. WebQSP represents a regime where the set of valid paths is comparatively small; CWQ represents the regime where valid path sets grow large and the difficulty of identifying the correct path among structurally valid alternatives is highest.

2.4.4 Multi-Hop Complexity

The combinatorial difficulty of multi-hop KGQA increases sharply with hop depth. For an entity with average out-degree d in the KG, the number of valid paths of length L reachable from the question entities grows as $|\mathcal{E}_q| \times d^L$. For well-connected Freebase entities where $d \approx 20$ and a question with up to three question entities, a three-hop question can produce up to 24,000 structurally valid paths, of which exactly one leads to the correct answer (He *et al.*, 2021b; Lan *et al.*, 2022b). This combinatorial explosion has concrete consequences for any approach that must discriminate among candidate paths: at each generation step, the model must select the correct next token from among a very large set of valid-but-incorrect alternatives, relying on its parametric knowledge to do so.

2.5 Constrained Decoding for Faithful Text Generation

Constrained decoding is the paradigm in which generation is restricted to a predefined valid set at each step through direct intervention in the decoding mechanism. Unlike prompting, which modifies the input to the LLM, constrained decoding modifies the token selection process itself: invalid tokens are assigned zero probability before sampling, making their generation structurally impossible.

2.5.1 Logit Masking

The foundational mechanism of constrained decoding is logit masking, described formally in Geng *et al.* (2023) and Willard and Louf (2023). A constraint oracle computes a binary validity mask over the vocabulary at each step; the mask is applied element-wise to the raw logit vector before softmax normalization, setting invalid token logits to negative infinity and thereby assigning them zero probability. Token selection then operates exclusively over the valid subset. The central design question in any constrained decoding system is how the validity mask is computed: what information the oracle uses, and how it changes across generation steps.

2.5.2 Grammar-Constrained Decoding

Geng *et al.* (2023) introduce Grammar-Constrained Decoding (GCD), which constrains generation to strings conforming to a context-free grammar, using an Earley parser as the

oracle. The valid next-token set at each step is the set of tokens that extend the current partial string as a valid grammar prefix. GCD additionally introduces input-dependent grammars, in which the grammar is parameterized by the input question, representing a partial step toward context-sensitive constraints. Willard and Louf (2023) address the computational cost of constrained decoding with an efficient finite-state machine formulation. By precomputing an index from automaton states to valid vocabulary subsets, the valid token set can be retrieved in amortized constant time at each generation step, making constrained decoding practical at inference speed.

The shared limitation of these format-constraint approaches is that they enforce structural validity of the output format (such as grammatical correctness or schema conformance) without enforcing factual correctness. A grammatically valid output can contain entirely hallucinated facts. The transition from format constraints to factual constraints requires a constraint oracle that verifies tokens against a knowledge source rather than against a grammar.

2.5.3 Knowledge Graph-Constrained Decoding

The application of constrained decoding to knowledge graph faithfulness is a more recent development. The core idea is to use the KG itself as the constraint oracle: at each generation step, only tokens that correspond to valid continuations of a KG reasoning path are admitted. This approach guarantees structural faithfulness, meaning every generated token corresponds to a verified KG triplet, which makes hallucination of KG facts structurally impossible.

Graph-Constrained Reasoning (GCR) (Luo *et al.*, 2025a) is the state of the art in this line of work. GCR constructs a KG-Trie (a prefix tree over all KG paths within L hops of the question entities, built by breadth-first search before generation begins) and uses it as the constraint oracle during beam-search decoding. The trie is queried at each generation step to return valid next tokens. GCR achieves 92.6 Hits@1 on WebQSP and 75.8 on CWQ with verified 100% structural faithfulness using a Llama-3.1-8B backbone. A key property is that the trie is constructed once before generation and never modified: the valid token set is fixed regardless of what has been generated.

Decoding on Graphs (DoG) (Li *et al.*, 2025) introduces step-wise trie expansion as a partial remedy to GCR's static oracle. DoG defines *well-formed chains* as sequences of KG triplets

in which every triplet exists in the KG and every entity is connected to the preceding triplet or is a question entity. It enforces this definition through step-wise expansion: after committing to an entity, the constraint trie is extended with all KG neighbors of that entity. This makes the oracle topologically dynamic, growing as reasoning progresses through the graph. DoG achieves 90.99 Hits@1 on WebQSP and 76.08 on CWQ with 100% structural faithfulness, and is training-free, requiring no fine-tuning. However, the expansion rule admits all KG neighbors of each visited entity regardless of their relevance to the question or the current reasoning direction; and per-step trie reconstruction sacrifices the efficiency of precomputed constraint structures.

RouterKGQA (Yuan *et al.*, 2026) takes a different approach, routing between a specialised KG-reasoning model and a general LLM based on question complexity. It applies semantic filtering using SBERT embeddings (specifically nomic-embed-text-v1) to select among candidate answers after generation, achieving competitive accuracy with reduced inference cost. RouterKGQA's semantic scoring operates at the answer selection level rather than at the token constraint level: paths are generated first and filtered retrospectively, rather than being constrained during generation.

2.5.4 Beam Search and Trie-Based Constraints

To understand how knowledge graph constraints are practically enforced during generation, it is essential to examine the two primary search algorithms and data structures involved: beam search and the prefix trie.

Beam Search. First introduced in the context of the HARPY speech recognition system (Lowerre, 1976), beam search is a heuristic search algorithm used in natural language processing to decode sequences from probabilistic models. Unlike greedy decoding, which exclusively selects the single most probable token at each step and risks generating sub-optimal global sequences, beam search maintains a set of the k most promising partial hypotheses (or "beams") at any given time. At each generation step, the algorithm expands all k current hypotheses by scoring their possible continuations, ranks the resulting sequences based on their cumulative log-probability, and prunes the search space back down to the top k candidates (Holtzman *et al.*, 2020). This approach mitigates the risk of early compounding errors while remaining computationally tractable, making it the dominant decoding strategy

for tasks requiring structured and coherent output.

The Prefix Trie. A trie, originating from the concept of a “trie memory" (retrieval) introduced by Fredkin (1960), is a deterministic tree-based data structure used to store a dynamic set of sequences, typically strings. In a trie, no single node stores a complete string; instead, a node’s position in the tree defines the exact prefix it represents. The branches extending from any given node represent the deterministic set of valid next characters or tokens. In the domain of constrained decoding, a pre-computed trie serves as a highly efficient constraint oracle. Given the current generated sequence prefix, traversing the trie to the corresponding node yields the exact subset of valid next tokens in amortised constant time, $O(1)$, relative to vocabulary size. This operation is critically more efficient than running continuous dynamic parser evaluations, making trie lookups the most scalable pattern for enforcing hard factual constraints at inference speed (Willard and Louf, 2023).

The KG-Trie. For knowledge graph-constrained generation, architectures such as Graph-Constrained Reasoning (GCR) utilize a specialized adaptation known as the KG-Trie (Luo *et al.*, 2025a). A KG-Trie serializes structurally verified multi-hop paths from the knowledge graph into string formats and maps them across the language model’s precise token vocabulary. Rather than nodes representing single alphabetical characters, each edge in the KG-Trie represents a specific valid token in the LLM’s vast vocabulary space.

When beam search is merged with a KG-Trie oracle (as visualized in Figure 2.2), the decoding process transforms into a strictly bounded traversal of the prefix tree. At each autoregressive step, the current beam hypotheses map to specific states in the KG-Trie. The language model calculates a probability distribution over its entire vocabulary, but a mathematical mask forces the probability of any candidate token not validly present as a child node in the trie to exactly zero ($-\infty$ prior to softmax normalization). The beam search then advances strictly down the permissible branches, functionally restricting the LLM’s autoregressive generation to a set of pre-verified deterministic paths while still allowing the model to express internal preference (via log-probabilities) among those factually valid chains.

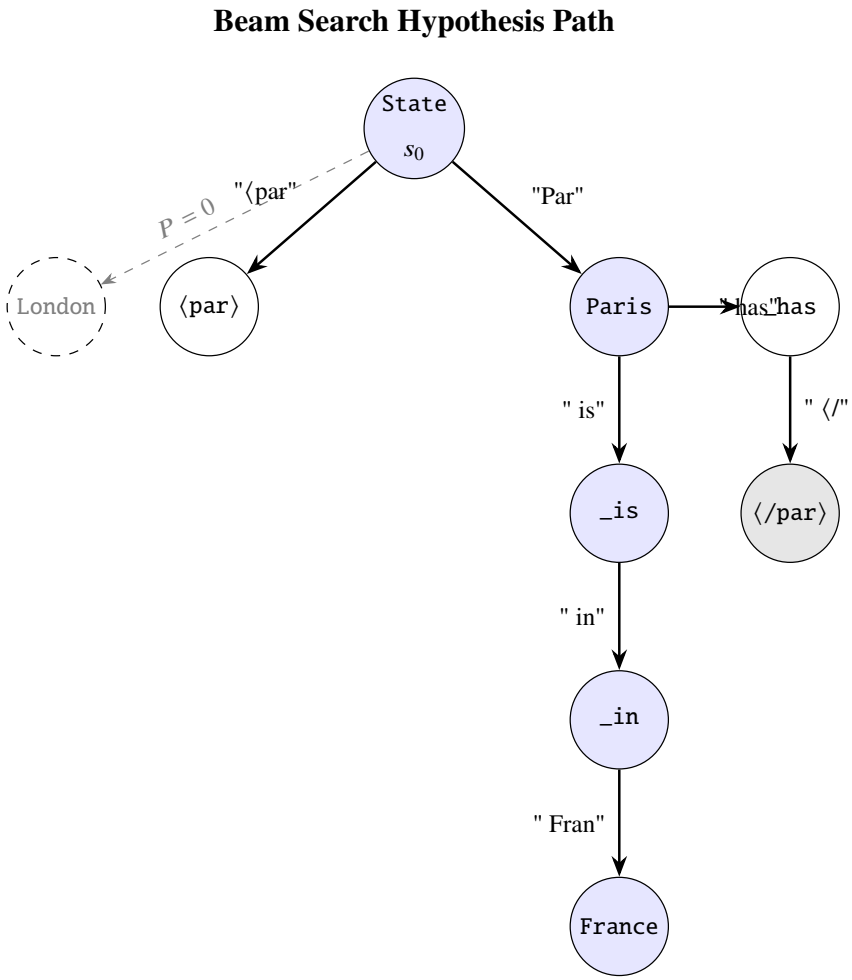

Figure 2.2 Visualization of KG-Trie Constrained Generation.

2.6 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG), introduced by Lewis *et al.* (2020), is the dominant paradigm for grounding LLM outputs in external knowledge. A dense retriever selects relevant passages from an external source based on embedding similarity, and a reader LLM generates the answer conditioned on the retrieved context and the question. Fusion-in-Decoder (Izacard and Grave, 2021) extends this by encoding multiple retrieved passages independently and fusing them in the decoder, improving coverage.

2.6.1 KG-Augmented Retrieval

Several works augment RAG with KG-structured context. He *et al.* (2021b) retrieve KG subgraphs and use attention over retrieved facts to guide multi-hop reasoning. Yasunaga *et al.* (2021) combine retrieved text passages and KG neighborhoods via graph neural networks.

Shi *et al.* (2021) retrieve KG paths as natural language context strings, providing path-shaped input to the reader. G-Retriever (He *et al.*, 2024) applies RAG specifically to textual graph understanding, retrieving graph-structured context for QA tasks.

2.6.2 Advanced RAG Variants

More recent RAG variants address the timing and precision of retrieval. Self-RAG (Asai *et al.*, 2024) inserts reflection tokens that allow the model to dynamically decide when retrieval is needed and whether retrieved passages support the claim being generated, improving retrieval precision and citation accuracy. FLARE (Jiang *et al.*, 2023b) uses the model's next-token prediction confidence as a retrieval trigger, retrieving fresh context when confidence falls below a threshold. IRCoT (Trivedi *et al.*, 2023) interleaves retrieval with chain-of-thought steps, retrieving relevant context at each intermediate reasoning step and improving multi-hop accuracy substantially. GraphRAG (Edge *et al.*, 2024) constructs community-structured knowledge graphs from document corpora and retrieves community summaries for complex relational queries, explicitly leveraging graph structure for improved coverage.

2.6.3 The Structural Limitation of Soft Grounding

Despite their empirical improvements, all RAG approaches share a structural limitation: retrieved context is a soft influence on generation. The retriever determines what the LLM sees; it does not determine what the LLM is permitted to generate. Because the LLM's generation mechanism remains unchanged (the distribution over the full vocabulary is computed autoregressively), there exists a nonzero probability of generating any token at any step, including tokens that correspond to no verified KG fact. Structural faithfulness with probability one, which guarantees that every generated token corresponds to a verified fact, cannot be achieved under any architecture that modifies only the input context without modifying the token selection mechanism (Luo *et al.*, 2025a; Geng *et al.*, 2023).

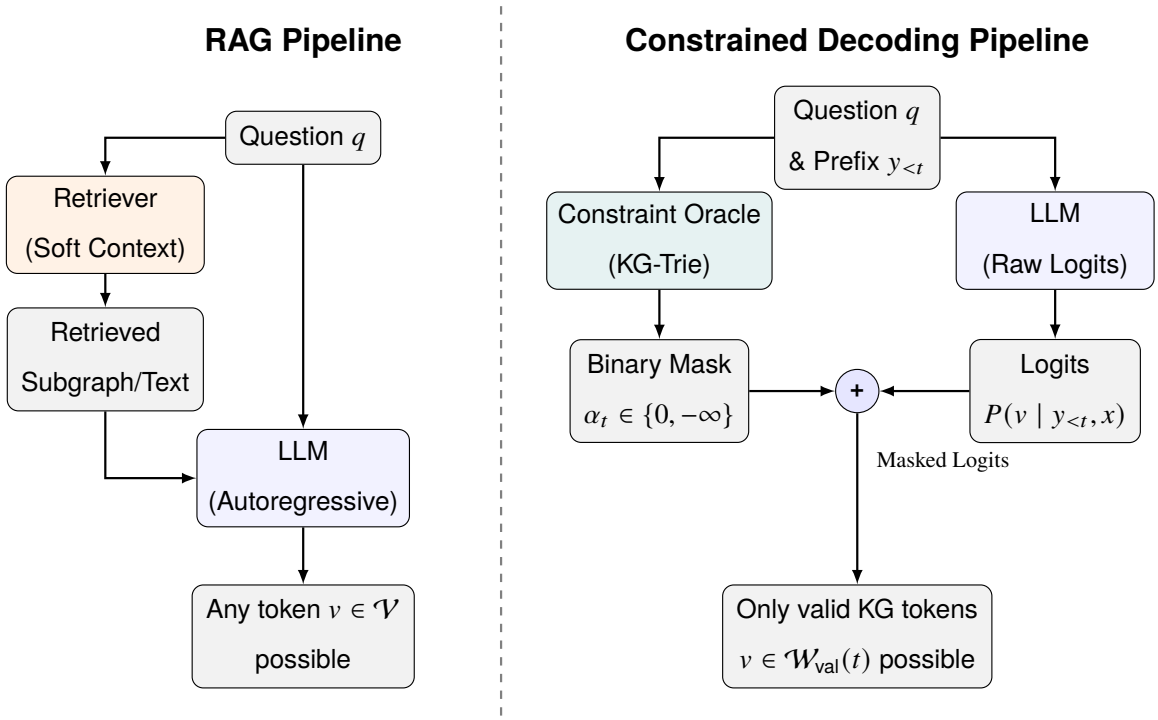

Figure 2.3 Architectural Comparison of Standard Retrieval-Augmented Generation (left) and KG-Constrained Decoding (right).

2.7 Semantic Relevance Scoring with Sentence Transformers

2.7.1 Sentence Transformer Embeddings

Sentence-BERT (Reimers and Gurevych, 2019) introduced a class of sentence encoders trained to produce dense vector representations in which semantically similar sentences cluster together and semantically dissimilar sentences are pushed apart. The SBERT architecture applies mean pooling over token representations from a BERT-like transformer encoder, producing a fixed-length embedding. Training uses a siamese network on labelled sentence pairs with a classification objective over entailment, neutral, and contradiction labels. The resulting embedding space supports cosine similarity as a reliable proxy for semantic alignment between sentences, without requiring task-specific fine-tuning at inference time.

2.7.2 Applications in Retrieval and Graph Reasoning

Sentence transformer similarity has been applied extensively in dense retrieval settings. Karpukhin *et al.* (2020) demonstrate that a bi-encoder architecture trained on question-passage pairs produces embeddings that outperform sparse retrieval (BM25) on open-domain QA. Xiong *et al.* (2021) refine bi-encoder training, improving coverage of relevant passages.

In the KGQA context, Shi *et al.* (2021) and Saxena, Chakrabarti and Talukdar (2022) apply semantic similarity to rank candidate KG paths for retrieval, selecting high-scoring paths as soft context for the reader LLM. These approaches use cosine similarity as a soft ranking signal over candidate paths, selecting the top-k for context provision.

2.7.3 Encoder Architecture Trade-offs

The choice of sentence encoder involves a quality-latency tradeoff that is particularly stringent in step-wise constrained decoding, where the encoder is invoked at each generation step. Larger encoders such as all-mpnet-base-v2 (110M parameters) and BGE-M3 (Chen *et al.*, 2024) (568M parameters) achieve higher performance on semantic textual similarity benchmarks but incur substantially higher inference latency. Lexical approaches such as BM25 are fast but cannot capture semantic similarity between paraphrase-equivalent paths that share no surface tokens. The all-MiniLM-L6-v2 model (Reimers and Gurevych, 2019), at 22M parameters with approximately 80ms per-query CPU latency, provides the optimal balance of semantic quality and inference speed for iterative step-wise applications.

2.8 Related Works

Building on the foundational concepts of constrained decoding, retrieval-augmented generation, and semantic scoring, this section steps back to survey the broader landscape of knowledge-augmented reasoning. To illustrate the evolution of the field, the surrounding literature is grouped into thematic clusters: LLM reasoning enhanced by knowledge graph access, structured retrieval-based question answering, and format-constrained generation. Examining these groups helps clarify both the significant achievements of prior methods and the limitations they share.

2.8.1 Semantic Parsing and Early Structured KGQA

Before the widespread adoption of large language models, classical approaches primarily tackled KGQA through semantic parsing or structured representation learning. There are two mainstream approaches corresponding to this era: parsing the question into executable KG queries like SPARQL, and grounding question and KG representations into a common space for reasoning.

In the first category, early semantic parsing methods relied on pre-defined question templates to synthesize queries, requiring strong domain knowledge (Berant *et al.*, 2013; Reddy, Lapata and Steedman, 2014; Bast and Haussmann, 2015). More recent methods apply deep learning techniques to automatically generate these executable logical forms (Yih *et al.*, 2015; Lan and Jiang, 2020). While these symbolic approaches promise high interpretability and exact reasoning, they suffer from significant real-world bottlenecks: they require expensive ground-truth logical forms for supervision, which are costly to obtain, and their performance severely degrades when the KG has missing links. Furthermore, model-generated queries are frequently non-executable due to minor syntax or semantic mismatches with the underlying graph schema, yielding no answers and lacking graceful fallback mechanisms (Yu *et al.*, 2023; Luo *et al.*, 2025a).

To alleviate the need for rigid logical forms, the second category of methods grounds text and graph representations into a shared continuous space to reason over. Some match question representations directly to KG facts (Miller *et al.*, 2016; Xu *et al.*, 2019) or KG structural representations (Zhang *et al.*, 2018; Sen, Mavadia and Saffari, 2023). Prominent examples include GraftNet (Sun *et al.*, 2018), PullNet (Sun *et al.*, 2019), and NSM (He *et al.*, 2021c), which enhance reasoning through deeply embedded graph-based reasoners. Incomplete graphs have also drawn attention, where missing links are inferred using KG embeddings (Saxena, Tripathi and Talukdar, 2020). While these structured techniques successfully side-step hallucination by operating natively over verified data, their expressive capability is limited compared to the generative flexibility and sophisticated multi-step reasoning offered by modern LLMs.

2.8.2 Knowledge Graph-Enhanced LLM Reasoning

More recent advances focus on the joint use of KGs and LLMs for KGQA, leveraging the sophisticated capabilities of LLMs for enhanced reasoning while mitigating hallucinations and knowledge gaps. Foundational studies initially explored eliciting reasoning via prompting alone, such as Chain-of-Thought (CoT) (Bosma *et al.*, 2022), Plan-and-solve (Wang *et al.*, 2023b), DecomP (Khot *et al.*, 2023), Self-consistency (Wang *et al.*, 2023d), and Tree-of-Thought (ToT) (Cao *et al.*, 2023b). Various studies also focus on fine-tuning LLMs on reasoning tasks to improve their abilities (Hoffmann *et al.*, 2022), with efforts extending to reinforcement learning for complex reasoning processes (e.g., OpenAI o1). However, the

persistent risk of hallucinations and missing domain knowledge compromises the faithfulness of pure generative reasoning, prompting the field to augment LLMs directly with KGs.

A large body of work explores how LLMs can be enhanced with KG access to guide derivations. Several methods treat LLMs as agents interacting with environments or KGs, such as ReACT (Yao *et al.*, 2023), StructGPT (Jiang *et al.*, 2023a), and ToG (Sun *et al.*, 2024), which prompts the LLM to explore relevant facts hop-by-hop using beam search on the KG. Others retrieve subgraphs or paths for intermediate reasoning supervision: KD-CoT (Wang *et al.*, 2023a) and RR (He *et al.*, 2022) retrieve facts to guide generative plans, while Jin *et al.* (2024) propose Graph Chain-of-Thought (GraphCoT) to explicitly retrieve graph neighbourhood information at every intermediate CoT step.

Specialized retrievers and graph encoders have also been developed to bridge text and graph modalities. Approaches like GNN-RAG (Mavromatis and Karypis, 2025) and GFM-RAG (Luo *et al.*, 2025b) adopt graph neural networks to effectively capture structures and retrieve from KGs. Luo *et al.* (2024) introduce Reasoning on Graphs (RoG), a planning-retrieval-reasoning framework that retrieves reasoning paths from KGs to conduct faithful reasoning. StructLM (Zhuang *et al.*, 2024) fine-tunes generalist models over massive structured datasets to fundamentally augment grounding capabilities. DECAF (Yu *et al.*, 2023) combines semantic parsing and LLM reasoning to jointly generate answers. While these approaches push the boundary of performance, they remain heavily reliant on instruction-following abilities (challenging for compact models) or computationally expensive fine-tuning.

Instead of strictly retrieving limited facts, methods such as Decoding on Graphs (DOG) (Li *et al.*, 2025) employ a general graph-aware decoding approach that uses the entire question graph natively without external retrievers, thus enabling reasoning over complete knowledge. Tree-of-Traversals (Markowitz *et al.*, 2024) similarly offers a zero-shot algorithm motivated by ToT to explore alternative solution structures.

Despite these advancements, many of these approaches apply KG knowledge as an advisory rather than hard structural constraint. For instance, Jiang *et al.* (2022) propose UniKGQA, unifying retrieval and reasoning for multi-hop KGQA through a single model with a shared parameter space. By tying the retrieval and reasoning objectives to a common representation, UniKGQA achieves strong performance. The shared representation, however, means that

errors in retrieval directly bias reasoning and vice versa, with no architectural separation between the two failure modes.

Han *et al.* (2018) develop OpenKE, an open-source platform for knowledge embedding that provides entity and relation representations learned from KG structure. These embeddings underpin downstream KGQA systems by providing semantically meaningful initializations for entity and relation tokens. A key limitation is that embedding-based representations are trained on fixed KG snapshots and do not generalise to new entities or relations that appear after training. Moreover, embedding similarity does not constitute a hard faithfulness guarantee.

Zhang *et al.* (2022) propose a subgraph-based KGQA approach that retrieves a question-relevant subgraph and performs message passing over it to propagate relational evidence before the LLM generates an answer. This approach substantially reduces the search space by focusing on a localised subgraph, improving efficiency. Its limitation is that the subgraph boundary is determined heuristically based on entity proximity, and correct answer entities that are far from the question entity in graph distance may be excluded from the subgraph entirely, creating a recall ceiling that is invisible to the answering module.

Wang *et al.* (2021) present KEPLER, a model that jointly optimises knowledge embedding and masked language modelling objectives within a single transformer, integrating factual KG knowledge directly into LLM parameters during pretraining. KEPLER improves entity representations and reduces hallucination on entity-centric tasks compared to models without KG pretraining. The limitation is that KG knowledge is internalised as parametric memory rather than enforced at generation time: KEPLER-pretrained models remain subject to the same autoregressive generation mechanism and can still hallucinate facts, particularly for long-tail entities underrepresented in the KG-augmented pretraining corpus.

Sun *et al.* (2019) introduce PullNet, an iterative graph construction and reasoning framework for multi-hop QA over heterogeneous text-and-KG sources. PullNet iteratively pulls relevant entities and facts from both corpora, building a question-specific graph over which inference is performed. This iterative construction improves coverage over fixed-subgraph retrieval approaches. However, because the graph is constructed from both text and KG, it conflates verified KG facts with unverified textual claims, and the downstream reasoning module cannot

distinguish between the two. The faithfulness guarantee provided by the KG is therefore diluted by the presence of textual evidence with no formal verification status.

Lin, Yang and Zhang (2022) conduct an empirical rethinking of KG-augmented LLM reasoning, systematically comparing a range of KG integration strategies across multiple benchmarks. Their analysis reveals that naïve KG injection (concatenating retrieved facts into the LLM prompt) often underperforms fine-tuned models, and that the quality of KG evidence retrieval is the primary performance bottleneck across all strategies. This finding motivates more sophisticated retrieval mechanisms but does not address the structural faithfulness problem: even with oracle retrieval, the LLM remains free to ignore or contradict the provided facts during generation.

Mavromatis and Karypis (2022) propose ReaRev, a reasoning revision framework for KGQA that applies iterative reasoning and revision over retrieved KG evidence, using a graph neural network to propagate and revise entity scores across reasoning steps. ReaRev achieves competitive performance on CWQ by allowing the model to revise earlier incorrect entity assignments as more evidence accumulates. The fundamental limitation is that the revision procedure operates over soft entity scores rather than hard path commitments: the model does not generate a verifiable reasoning chain, and the final entity selection is made by argmax over a distribution rather than by following a path that can be independently verified against the KG.

These works collectively demonstrate the value of explicit KG access for improving LLM reasoning quality and interpretability. Their shared limitation is that KG access is advisory rather than constraining: the LLM attends to retrieved KG context but is free to generate beyond it. Across all approaches reviewed, the reasoning chains produced may be plausible and KG-informed without being guaranteed to correspond to verified KG paths. The absence of a hard structural guarantee is not merely a theoretical concern: Luo *et al.* (2025a) measure this gap directly, finding that even the strongest KG-enhanced approaches produce hallucinated reasoning steps in approximately one-third of cases, establishing a systematic upper bound on what soft KG grounding can achieve.

2.8.3 Retrieval-Based KGQA

Earlier KGQA approaches relied on structured retrieval over the knowledge graph rather than LLM generation. These methods avoid LLM hallucination by never invoking a free-form generative model, operating instead within the structured boundaries of the KG. The following works span the primary mechanisms in this family, from semantic parsing and embedding-based retrieval to graph neural network propagation and path-level supervision.

Yih *et al.* (2016) establish the WebQSP benchmark and introduce a semantic parsing approach that maps natural language questions to SPARQL queries executed against Freebase. By constraining the answer derivation to formally verified SPARQL execution, this approach provides exact symbolic faithfulness. Its core limitation is the brittleness of the parsing step: small surface variations in question phrasing can produce incorrect parses that yield structurally valid but semantically incorrect SPARQL queries. The approach also requires exhaustive annotation of SPARQL queries for training, which is costly and difficult to scale to complex compositional questions.

Talmor and Berant (2018) introduce ComplexWebQuestions and a corresponding compositional semantic parsing approach that decomposes complex questions into simpler sub-questions before generating SPARQL. This decomposition strategy extends the coverage of semantic parsing to compositional queries but does not resolve the brittleness problem: decomposition errors compound across sub-questions, and the final SPARQL query may be structurally valid while answering a subtly different question than the one posed.

He *et al.* (2021b) propose a multi-hop KGQA method that learns intermediate supervision signals to guide path selection through the KG, showing that explicit intermediate supervision at each hop substantially improves performance at greater reasoning depths. By training the model to predict correct intermediate entities, this approach reduces the combinatorial difficulty of multi-hop search. Its limitation is that intermediate supervision signals are derived from annotated answer paths, which are expensive to obtain and may not be available for every benchmark or KG. Moreover, the approach is constrained to a fixed maximum hop depth and does not adapt its reasoning depth dynamically based on question complexity.

Yasunaga *et al.* (2021) propose QA-GNN, combining retrieved text passages and KG

neighbourhood entities in a joint graph neural network that propagates information between passage token nodes and KG entity nodes. By jointly reasoning over both modalities, QA-GNN improves performance on questions that require combining textual and structured evidence. The joint graph conflates verified KG facts with unverified textual claims in the same propagation mechanism, meaning that noisy or incorrect text passages can corrupt entity scores derived from KG structure. The approach also requires constructing a joint graph at inference time for each question, incurring significant preprocessing overhead.

Shi *et al.* (2021) retrieve KG reasoning paths by computing semantic similarity between the question and natural language descriptions of candidate paths, providing the top-k highest-scoring paths as context strings to a reader LLM. This retrieval-plus-reading architecture separates the structured retrieval step from the natural language generation step. The limitation is that the ranking function uses soft cosine similarity over surface-form path descriptions, which may assign high scores to paths that are lexically similar to the question but semantically incorrect. The reader LLM, receiving paths as plain text, has no mechanism to verify that the provided paths are actual KG edges.

Saxena, Chakrabarti and Talukdar (2022) apply entity and relation embeddings to score and select multi-hop KG paths for KGQA without invoking an LLM generator. Entity and relation representations are learned from KG structure using embedding objectives, and multi-hop path scores are computed by composing relation embeddings. This approach is fully contained within the structured KG and avoids generative hallucination entirely. However, embedding-based path scoring cannot capture the full semantic complexity of natural language questions: compositional questions with negation, comparison, or conjunction are difficult to encode faithfully as compositions of relation embeddings, creating a coverage ceiling on complex question types.

Lan *et al.* (2022b) provide a comprehensive survey of KGQA approaches, systematically analysing the trade-offs between semantic parsing, embedding-based, and retrieval-based methods across benchmarks. Their analysis reveals that retrieval-based approaches consistently struggle with tail entities and uncommon relation types that are underrepresented in training data, and that no single retrieval mechanism dominates across all question types and hop depths. This survey motivates the turn toward LLM-based approaches as a more flexible alternative.

Sun *et al.* (2019) develop PullNet, which iteratively pulls relevant KG facts and text passages into a dynamically constructed evidence graph, performing inference over this graph to answer multi-hop questions. The iterative construction allows PullNet to retrieve evidence at multiple depths without fixing the hop depth in advance. Its limitation is that the evidence graph mixes KG and text sources without distinguishing their reliability, and the iterative pull procedure introduces substantial latency that scales with question complexity.

Das *et al.* (2018) introduce "Go for a Walk and Arrive at the Answer," an end-to-end differentiable path-finding approach that trains a policy to traverse the KG from a question entity to the answer entity. The policy is trained with reinforcement learning using answer correctness as reward. This approach directly models the reasoning process as graph traversal and produces interpretable paths. Its limitation is that reinforcement learning from sparse answer-correctness rewards is highly unstable during training, requiring careful reward shaping. The policy also generalises poorly to question types not well-represented in the training distribution.

Zhang *et al.* (2022) propose subgraph-based reasoning that extracts a question-relevant subgraph from the full KG and performs message passing over it before selecting the answer entity. By restricting reasoning to a local subgraph, this approach dramatically reduces the search space and inference cost. The subgraph boundary is determined by entity proximity to the question entity, which creates a recall ceiling: answer entities that are topologically distant from the question entity in the KG may be excluded from the subgraph even when they are the correct answer. This limitation is structurally linked to the hop-depth assumptions built into subgraph extraction heuristics.

These retrieval-based approaches avoid LLM hallucination by operating within the structured KG rather than generating freely. Their shared limitation is expressiveness: they are typically constrained to fixed path structures, fixed hop depths, or fixed retrieval budgets, and do not leverage the flexible multi-step reasoning capability of modern LLMs. The transition to LLM-based approaches in more recent work improved naturalness and coverage on complex compositional questions, but at the cost of the structural faithfulness guarantees that retrieval-based methods implicitly provide.

2.8.4 Format-Constrained and Entity-Constrained Generation

A distinct line of research addresses factual faithfulness by imposing hard constraints on what a generative model is permitted to produce, independently of the input context. Unlike retrieval-based approaches that restrict the information available to the model, these works modify the generation mechanism itself. The following cluster traces the development of this paradigm from early lexical constraints through grammar-based and automaton-based frameworks.

De Cao *et al.* (2021) introduce autoregressive entity retrieval (GENRE), constraining generation to a finite set of entity names using a trie constructed over all known entity surface forms. By restricting beam search to valid trie paths, GENRE guarantees that every generated token sequence corresponds to a real entity in a fixed vocabulary. This approach eliminates hallucination of entity names and achieves strong performance on entity disambiguation and linking tasks. Its limitation for KGQA is that constraining to entity names does not enforce path-level faithfulness: generating a real entity name at each step does not ensure that consecutive entities are connected by a verified KG edge. A sequence of individually valid entity names may constitute a reasoning chain that has no corresponding path in the KG.

Geng *et al.* (2023) generalise entity-level trie constraints to arbitrary context-free grammar constraints in Grammar-Constrained Decoding (GCD). GCD uses an Earley parser as the constraint oracle, admitting only tokens that extend the current partial string as a valid grammar prefix. GCD additionally introduces input-dependent grammars parameterised by the input question, allowing the valid token set to depend on question content. GCD establishes that grammar constraints are computationally tractable and empirically improve structured output quality across multiple tasks. The limitation is that a context-free grammar can enforce syntactic or format correctness but cannot encode factual membership in a knowledge graph: a string may conform perfectly to a grammar specification while containing entirely hallucinated entities or relations.

Willard and Louf (2023) address the computational cost of grammar-constrained decoding with an efficient finite-state machine (FSM) formulation. By precomputing an index from FSM states to valid vocabulary subsets, the valid token set can be retrieved in amortised constant time at each generation step, making constrained decoding practical at inference

speed for production systems. This formulation reduces the per-step constraint overhead from the Earley parsing cost (which is cubic in input length) to effectively $O(1)$ amortised lookup. The limitation remains the same as GCD: the FSM encodes structural or format correctness, not factual correctness. An FSM-constrained system can still generate structurally valid outputs that are factually incorrect with respect to any external knowledge source.

Deutsch, Upadhyay and Roth (2019) propose a general algorithm for constrained sequential inference applicable across any structured prediction task, framing constraint enforcement as inference in a product of finite-state automata. This framework unifies a broad range of constraint types under a common algorithmic schema and demonstrates that hard constraint enforcement is compatible with a variety of sequence models. Its limitation for KGQA is that the constraints it enforces are defined over output string structure rather than over external knowledge: the automata encode properties of the output form, not membership in a KG edge set.

Poesia *et al.* (2022) introduce Synchromesh, a constrained generation system for program synthesis that uses a completion engine to enforce syntactic validity of generated code. By grounding constraint enforcement in a language-specific parser, Synchromesh ensures that all generated programs are syntactically valid. This work establishes that constraint oracles can be derived from formal language specifications beyond context-free grammars. Applied to KGQA, an analogous approach would enforce syntactic validity of SPARQL queries; however, a syntactically valid SPARQL query may still reference entities or relations absent from the target KG, producing an empty result set rather than a faithfully grounded answer.

Scholak, Schucher and Bahdanau (2021) present PICARD, a method for parsing SQL queries with constrained autoregressive inference. PICARD uses an incremental parser to reject tokens that would make the partial output unparseable or semantically invalid with respect to a database schema, thereby constraining generation to schema-valid SQL. PICARD demonstrates that schema-grounded constraints substantially reduce invalid query generation in text-to-SQL tasks. The limitation for KG settings is that SQL schema constraints enforce structural consistency with a fixed schema, not factual path traversal through a dynamic relational graph. A schema-valid query may still produce an incorrect answer if the underlying reasoning is unsound.

Anderson *et al.* (2017) introduce Guided Open Vocabulary Image Captioning (GOVIC), an early demonstration of constrained beam search for lexically-constrained generation in image captioning, enforcing inclusion of specified words in the output. While the domain differs from KGQA, this work establishes constrained beam search as a practical mechanism for enforcing lexical constraints during generation and demonstrates the feasibility of hard inclusion constraints. Its limitation is that lexical constraints enforce presence of specific surface forms without enforcing their factual correctness or logical coherence with the rest of the generated output.

Hokamp and Liu (2017) introduce constrained decoding via a grid beam search algorithm that enforces specified lexical constraints in neural machine translation. Grid beam search maintains separate hypothesis sets for each subset of satisfied constraints and merges them to produce outputs satisfying all constraints. This algorithm demonstrates that complex constraint sets can be enforced in polynomial time during beam search. The limitation is that the constraint set must be specified in advance and is fixed throughout generation; there is no mechanism for the constraint oracle to consult an external knowledge source at each step to determine which tokens are valid continuations.

Lu *et al.* (2021) propose NeuroLogic Decoding, a decoding algorithm that enforces propositional logic constraints over lexical items during generation, extending the class of enforceable constraints beyond simple inclusion to disjunctions, conjunctions, and negations. NeuroLogic demonstrates that logically complex constraints can be enforced efficiently during beam search without prohibitive computational overhead. For KGQA applications, the limitation is that propositional logic constraints over lexical items still operate at the surface-form level: a NeuroLogic constraint can require that a specific entity name appears in the output, but cannot verify that this entity is connected to the preceding entity by a verified KG edge.

Zhang *et al.* (2023) analyse the computational tractability of constrained decoding under various constraint classes, establishing theoretical bounds on the complexity of constraint enforcement as a function of grammar type and vocabulary size. Their analysis confirms that finite-state constraints admit efficient enforcement via precomputed indices, while context-sensitive constraints require per-step evaluation that may be prohibitively slow. This theoretical result directly motivates the design choice in KG-constrained systems of representing valid paths as tries (finite-state structures) rather than as arbitrary graph reachability queries.

These works collectively establish that hard constraints on generation are computationally tractable and empirically beneficial across multiple domains. Their shared limitation for KGQA is that they enforce output *format* rather than *factual content*: constraining generation to grammatically valid strings, schema-valid queries, or lexically specified surface forms does not ensure that the generated strings correspond to verified KG paths. The extension from format constraints to KG path constraints requires a constraint oracle that encodes the relational structure of the knowledge graph itself — mapping valid token sequences not to grammar derivations but to traversable edges in a specific, dynamic knowledge base.

2.8.5 Semantic Similarity in Graph Reasoning

Several works apply semantic similarity signals to guide reasoning over knowledge graphs. Shi *et al.* (2021) rank candidate KG paths by semantic similarity between the question and path descriptions, retrieving high-scoring paths as evidence. Saxena, Chakrabarti and Talukdar (2022) learn embeddings for KG entities and relations that support similarity-based multi-hop path selection. He *et al.* (2021b) use attention-guided traversal over retrieved KG subgraphs, with attention weights reflecting the relevance of each subgraph region to the question. PathRetriever (Asai *et al.*, 2020) learns to retrieve reasoning paths from evidence graphs by scoring paths using a recurrent model over their node sequences.

These works demonstrate that semantic similarity can effectively differentiate relevant from irrelevant KG paths in a retrieval setting. The key architectural distinction is that semantic similarity is applied as a *soft ranking signal*: candidate paths are scored and the highest-scoring ones are provided to the LLM as context or used to select among generated candidates. No existing work applies semantic similarity as a *hard binary admission criterion* that determines which tokens the LLM is permitted to generate at each decoding step. The distinction between soft retrieval signals and hard generation constraints is significant: the former influences what the LLM sees, while the latter determines what the LLM can produce.

2.8.6 Hallucination Mitigation in Knowledge-Intensive Tasks

Beyond the constrained decoding paradigm, a range of approaches attempt to reduce hallucination in knowledge-intensive tasks. Retrieval-augmented approaches (including RAG (Lewis *et al.*, 2020), Self-RAG (Asai *et al.*, 2024), FLARE (Jiang *et al.*, 2023b),

IRCoT (Trivedi *et al.*, 2023), and GraphRAG (Edge *et al.*, 2024)) address the problem by providing relevant external context, improving factual coverage without modifying the generation mechanism. Wang *et al.* (2023c) improve consistency by majority-voting over multiple CoT chains. Huang *et al.* (2024) study whether LLMs can self-correct reasoning errors, finding that self-correction is generally ineffective without external feedback, reinforcing that internal parametric knowledge cannot reliably detect its own errors.

Post-hoc verification approaches (Nakano *et al.*, 2022; Gao *et al.*, 2023) check generated claims against external sources after generation and revise or flag unsupported statements. While effective at identifying errors, these approaches require a separate verification pass and do not prevent hallucination during generation. Agrawal *et al.* (2024) systematically analyse failure modes in RAG systems, finding that even with relevant context provided, LLMs regularly produce statements unsupported by the retrieved evidence.

2.8.7 Concurrent Work

RouterKGQA (Yuan *et al.*, 2026) is the work most closely related to this research in terms of shared components. RouterKGQA routes KGQA queries between a specialised KG reasoning model and a general LLM based on question complexity, and applies SBERT-based semantic similarity (using the nomic-embed-text-v1 encoder) to filter candidate reasoning paths. It achieves competitive performance with reduced inference cost, averaging approximately 1.15 LLM calls per question.

The key architectural distinction is the level at which semantic scoring is applied. RouterKGQA scores complete reasoning paths after they have been generated, using similarity to filter among candidates retrospectively. This approach does not modify the token selection mechanism during generation: the LLM generates freely, and semantic similarity is applied post-hoc to select among the outputs. By contrast, this thesis uses semantic similarity as a hard admission criterion at the token-constraint level. In this design, similarity determines which tokens the LLM is permitted to generate at each decoding step, not merely which completed outputs are retained afterward. This is a qualitatively different architectural intervention that RouterKGQA does not explore. RouterKGQA also does not provide a metric for constraint permissiveness and does not measure the degree to which its candidate sets contain structurally valid but semantically irrelevant paths.

2.9 Synthesis: Three Unresolved Gaps

The preceding review reveals that two distinct research trajectories (constrained decoding for structural faithfulness, and semantic similarity for relevance-guided reasoning) have developed largely in parallel without being integrated at the constraint-oracle level. Constrained decoding approaches (Luo *et al.*, 2025a; Li *et al.*, 2025) achieve structural faithfulness but construct their constraint oracles without reference to question semantics or generation context, producing valid token sets that contain large numbers of irrelevant paths. Semantic similarity approaches (Shi *et al.*, 2021; Saxena, Chakrabarti and Talukdar, 2022; Yuan *et al.*, 2026) improve relevance but operate as soft signals that influence retrieval or post-hoc selection without modifying the generation mechanism.

Three specific gaps emerge from this review.

Gap 1 – The Permissiveness Problem. Existing KG-constrained decoding frameworks construct their constraint oracles as functions of graph structure and question entities alone, without reference to question semantics or the evolving generation state. For multi-hop questions, this produces valid token sets containing thousands of structurally correct but semantically irrelevant paths at every generation step. The LLM must discriminate among these paths using its own parametric knowledge, reintroducing the hallucination risk that constrained decoding was designed to eliminate (Luo *et al.*, 2025a; Li *et al.*, 2025).

Gap 2 – The Static-Dynamic Efficiency Tradeoff. GCR's precomputed static trie is computationally efficient but semantically blind and identical at every generation step. DoG's step-wise topological expansion introduces dynamism at the cost of per-step trie reconstruction and without semantic filtering (Li *et al.*, 2025; Willard and Louf, 2023). No existing framework achieves semantic responsiveness (where the valid set narrows progressively as reasoning commits to a specific direction) while preserving the efficiency of precomputed constraint structures.

Gap 3 – The Absence of a Constraint Quality Metric. No existing work provides a metric that quantifies constraint permissiveness independently of downstream answer accuracy. Without such a metric, it is impossible to measure progress on reducing irrelevant paths in the valid set, compare frameworks on constraint quality independently of model quality, or

diagnose permissiveness as a failure mode in deployed systems (Luo *et al.*, 2025a; Li *et al.*, 2025).

Table 2.1 Summary of Key Related Works and Gap Coverage.

Work	**Approach**	**Structural faithfulness**	**Semantic conditioning**
Luo *et al.* (2025a)	Static KG-Trie decoding	Yes (100%)	No
Li *et al.* (2025)	Step-wise topological expansion	Yes (100%)	No
Yuan *et al.* (2026)	Routing + post-hoc filtering	Probabilistic	Yes (answer-level)
Luo *et al.* (2024)	Reasoning path planning	No	Partial
Jiang *et al.* (2023a)	LLM over structured data	No	Partial
Jin *et al.* (2024)	Graph-augmented CoT	No	Partial
Jiang *et al.* (2022)	Unified retrieval and reasoning	No	Partial
Wang *et al.* (2021)	KG-pretrained language model	No	Partial
Mavromatis and Karypis (2022)	Iterative reasoning revision	No	Partial
He *et al.* (2021b)	KG subgraph attention	No	Partial
Yasunaga *et al.* (2021)	Text-KG graph neural network	No	Partial
Das *et al.* (2018)	RL-based KG path traversal	No	Soft
Saxena, Chakrabarti and Talukdar (2022)	Embedding-based path selection	No	Soft
Shi *et al.* (2021)	Path retrieval by similarity	No	Soft (retrieval)
Asai *et al.* (2024)	Retrieval with reflection	No	Soft
Edge *et al.* (2024)	Community-structured retrieval	No	Soft
Geng *et al.* (2023)	Grammar-constrained decoding	Format only	No
Willard and Louf (2023)	FSM-indexed constrained decoding	Format only	No
De Cao *et al.* (2021)	Entity trie-constrained decoding	Entity names only	No
Scholak, Schucher and Bahdanau (2021)	Schema-constrained SQL parsing	Schema only	No
Lu *et al.* (2021)	Logic-constrained decoding	Format only	No

These three gaps collectively identify an open problem at the intersection of the two research trajectories reviewed in this chapter: integrating semantic relevance scoring directly into the constraint oracle, conditioning the valid token set jointly on the knowledge graph, the question semantics, and the evolving generation state.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter outlines the methodological framework for the Dynamic Context-Aware Trie (DCA-Trie), a system designed to address the permissiveness problem in graph-constrained decoding. To evaluate whether DCA-Trie successfully balances structural faithfulness with contextual relevance, this chapter introduces a diagnostic metric: the Semantic Irrelevance Ratio (SIR). After formalizing the core problem and this SIR metric, the discussion details the underlying semantic scoring mechanism and the architectural design of both DCA-Trie variants. Finally, it establishes the baseline systems, evaluation protocol, and specific ablation strategies used to isolate the framework's effects, setting the boundaries for the experimental results analysed in Chapter 4.

3.2 Formal Problem Specification

3.2.1 Restating the Core Limitation of Existing Frameworks

As established in Chapter 2, current graph-constrained frameworks (notably GCR (Luo *et al.*, 2025a) and DoG (Li *et al.*, 2025)) rely on a deterministic oracle that looks only at the fixed graph $\mathcal{G}$ and the starting entities $\mathcal{E}_q$. Regardless of whether the system builds this oracle before decoding or expands it on the fly, the set of valid tokens at any step t remains locked to structural logic:

$$\mathcal{W}_{\text{val}}^{\text{GCR}}(t) = f(\mathcal{G}, \mathcal{E}_q) \quad \forall\, t \tag{3.1}$$

By ignoring both the semantic intent of the question q and the reasoning trajectory already captured in the prefix $y_{<t}$, this formulation treats every structurally reachable path as equally plausible. In dense, multi-hop Freebase environments, this blindness causes catastrophic search space explosion. The framework might admit over 8,000 paths at depth 3, even though only a single path logically answers the query.

The core issue is that these systems conflate two separate requirements. *Structural validity*

ensures the LLM only generates tokens that correspond to real graph edges, which current models do well. *Contextual relevance* ensures the LLM only generates tokens that logically connect to the specific question at hand, which current models fail to do. This gap between structural compliance and semantic usefulness is the *permissiveness problem*.

3.2.2 The Upgraded Oracle Specification

DCA-Trie addresses this problem by defining an upgraded constraint oracle that combines structural validity with semantic context. The oracle conditions on both the input question and the evolving generation prefix:

$$\mathcal{W}_{\text{val}}^{\text{DCA}}(t) = f(\mathcal{G},\, \mathcal{E}_q,\, q,\, y_{<t}) \tag{3.2}$$

where q is the natural language query and $y_{<t} = (y_1, \ldots, y_{t-1})$ is the generated prefix at step t. To be methodologically sound, the upgraded oracle must satisfy three conditions aligned with the project objectives in Chapter 1:

1. **Structural Faithfulness (Condition F):** Every candidate token $v \in \mathcal{W}_{\text{val}}^{\text{DCA}}(t)$ must correspond to a valid prefix of a path $p \in \mathcal{G}$ at every step t. This preserves the 100% structural faithfulness guarantee achieved by GCR and DoG and prevents ungrounded tokens from entering the valid set.

2. **Semantic Relevance Reduction (Condition R):** The Semantic Irrelevance Ratio (SIR) under DCA-Trie must be lower than the corresponding SIR under GCR when averaged over the evaluation corpus. Using SIR as defined in Section 3.4:

$$\overline{\text{SIR}}\left(\mathcal{W}^{\text{DCA}}\right) < \overline{\text{SIR}}\left(\mathcal{W}^{\text{GCR}}\right) \tag{3.3}$$

3. **Recall Preservation (Condition P):** Semantic pruning must not remove gold paths excessively. Operationally, the false negative rate (FNR) on the WebQSP validation split must remain below 5%:

$$\text{FNR} = \frac{|\{q \in \mathcal{Q}_{\text{val}} : p_q^* \notin \mathcal{W}_{\text{val}}^{\text{DCA}}(t)\}|}{|\mathcal{Q}_{\text{val}}|} < 0.05 \tag{3.4}$$

where p_q^* is the gold path for question q and Q_{val} is the held-out 100-question validation set.

3.3 System Architecture Overview

Because DCA-Trie explicitly targets the constraint oracle layer, its architecture builds directly on the baseline pipeline from Graph-Constrained Reasoning (GCR) (Luo *et al.*, 2025a). This reuse isolates the experimental effects: by holding the rest of the generation pipeline constant, the system guarantees any improvement in latency or search efficiency comes solely from the new oracle mechanism. The four layers function as follows:

1. **Entity Linking Layer (Unmodified):** A named entity recognition tool first isolates the core query entities $\mathcal{E}_q$ from the raw text question q. These isolated entities dictate where the downstream search algorithm is allowed to begin exploring the knowledge graph.

2. **Constraint Oracle Layer (DCA-Trie Contribution):** This is the mechanism that governs which tokens the LLM is allowed to generate. In GCR, this layer builds a static prefix tree (a KG-Trie) containing every possible blind path reachable within L hops of the starting entities. DCA-Trie completely overhauls this. Whether via static pre-filtering (v1) or dynamic, step-by-step expansion (v2), the new oracle ensures no structural edge is admitted into the trie unless it also proves its semantic relevance to the question.

3. **Constrained Decoding Layer (Unmodified):** During autoregressive generation using beam search, this layer intercepts the LLM logit distribution, applying a binary mask fetched from the oracle layer. By forcing the LLM to only output structurally and semantically approved tokens, it maintains 100% ground-truth faithfulness to the knowledge graph.

4. **Inductive Reasoning Layer (Unmodified):** Once the constrained beam search produces K top reasoning paths, GPT-4o-mini analyses these paths concurrently to synthesize the final natural language answer.

Figure 3.1 illustrates the layout, isolating the constraint oracle layer to highlight where the core semantic intervention occurs.

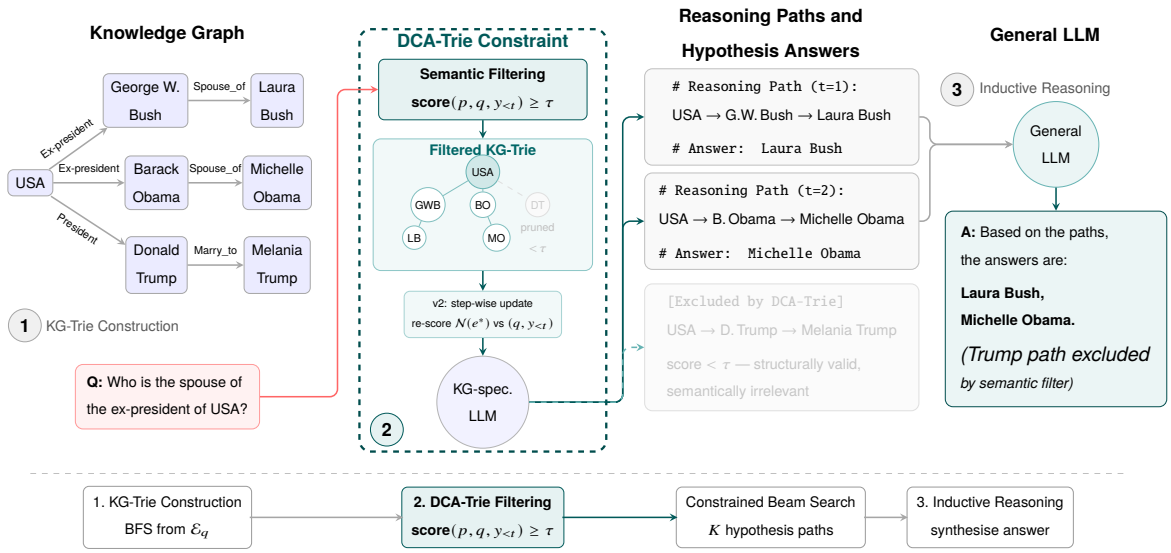

Figure 3.1 DCA-Trie System Architecture. The constraint oracle layer (teal) is the primary contribution; all other layers are inherited from GCR (Luo *et al.*, 2025a).

3.4 The Semantic Irrelevance Ratio: Definition and Measurement

3.4.1 Standard Metric Deficiencies

Existing performance markers for knowledge graph decoding (such as Hits@1, F1, and structural faithfulness ratios) all share a major limitation: they measure output, not process. While these metrics indicate whether a framework found the right answer and stayed inside the graph, they obscure how the framework reached that answer. Even if a model successfully resolves the correct gold path, it might do so while blindly wandering through an enormous, resource-draining set of completely irrelevant paths.

Thus, verifying accuracy is insufficient; the constraint mechanism itself must be measured for computational focus. To quantify exactly how permissive the oracle is independently of final output accuracy, this thesis introduces a novel diagnostic metric: the Semantic Irrelevance Ratio (SIR).

3.4.2 Formal Definition

Suppose $\mathcal{P}(q,t)$ represents all paths that the constraint oracle allows at decoding step t for question q. To calculate SIR, the system must first define irrelevance. An individual candidate path $p \in \mathcal{P}(q,t)$ is structurally valid, but it fails the semantic stringency test if its contextual similarity drops below a hard baseline threshold τ_{ref}:

$$\text{irrelevant}(p, q, y_{<t}) = \mathbf{1}\Big[\cos\big(E(p),\ E(\text{concat}(q, y_{<t}))\big) < \tau_{\text{ref}}\Big] \tag{3.5}$$

where $E(\cdot)$ is the `all-MiniLM-L6-v2` sentence encoder, and $\tau_{\text{ref}} = 0.3$ is a fixed reference threshold. This reference threshold is intentionally non-tunable and separate from DCA-Trie's admission threshold τ.

The step-level SIR for question q at decoding step t is then defined as:

$$\text{SIR}(q,t) = \frac{\sum_{p \in \mathcal{P}(q,t)} \text{irrelevant}(p, q, y_{<t})}{|\mathcal{P}(q,t)|} \tag{3.6}$$

Corpus-level SIR is computed by averaging over all questions and all valid decoding steps up to depth L:

$$\overline{\text{SIR}} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{L_q} \sum_{t=1}^{L_q} \text{SIR}(q,t) \tag{3.7}$$

where L_q is the hop length of the generated reasoning chain for question q.

3.4.3 Properties and Interpretation

By definition, $\overline{\text{SIR}} \in [0, 1]$. A value near 1 indicates that most admitted paths are irrelevant, so the model still faces a large and noisy search space. A value near 0 indicates that the constraint set is tightly aligned with question semantics. Prior constrained frameworks such as GCR and DoG tend to show high SIR, especially at larger hop depths where path counts increase rapidly. DCA-Trie is designed to reduce SIR by filtering semantically weak paths while retaining structurally valid ones. In reporting, SIR is analysed by hop depth to evaluate how this effect scales with reasoning complexity.

3.5 Semantic Relevance Scoring Mechanism

3.5.1 Scorer Architecture

The semantic relevance scorer implements the transition from a static oracle $f(\mathcal{G}, \mathcal{E}_q)$ to a context-aware oracle $f(\mathcal{G}, \mathcal{E}_q, q, y_{<t})$. It takes a candidate KG path p, the question q, and the partial generation $y_{<t}$, and returns a relevance score in $[0, 1]$.

A path $p = (e_0, r_1, e_1, \ldots, r_L, e_L)$ is first serialized into a text sequence:

$$\mathrm{str}(p) = e_0 \parallel r_1 \parallel e_1 \parallel \cdots \parallel r_L \parallel e_L \tag{3.8}$$

where entity IDs are mapped to readable Freebase names when available, and relation strings are normalized (e.g., underscore replacement). The query-side context is serialized as:

$$\mathrm{query}(q, y_{<t}) = q \parallel \texttt{[SEP]} \parallel \mathrm{str}(y_{<t}) \tag{3.9}$$

where $\mathrm{str}(y_{<t})$ is the detokenised generation prefix.

Both $\mathrm{str}(p)$ and $\mathrm{query}(q, y_{<t})$ are encoded using `all-MiniLM-L6-v2` (Reimers and Gurevych, 2019) into 384-dimensional vectors. Relevance is computed as cosine similarity:

$$\mathrm{score}(p,\ q,\ y_{<t}) = \cos\big(E(\mathrm{str}(p))\,,\ \ E(\mathrm{query}(q, y_{<t}))\big) \tag{3.10}$$

A candidate path is admitted into $\mathcal{W}_{\mathrm{val}}^{\mathrm{DCA}}(t)$ if and only if $\mathrm{score}(p, q, y_{<t}) \geq \tau$, where τ is the admission threshold. This binary decision integrates the semantic score into the same logit-masking framework defined in Equation. The full scoring workflow is shown in Figure 3.2.

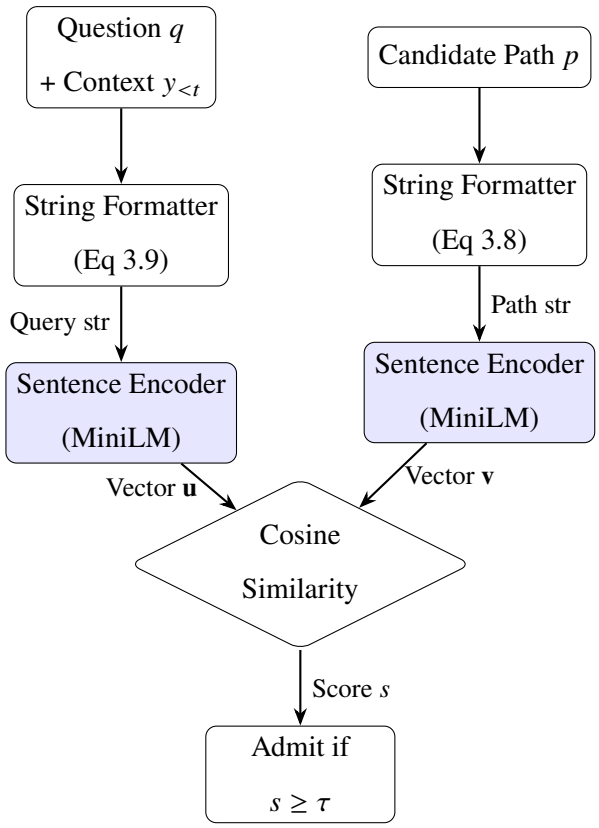

Figure 3.2 Semantic Relevance Scoring Mechanism.

3.5.2 Threshold Selection

The admission threshold τ is tuned on a separate 100-question WebQSP validation split. Selection prioritizes Condition P (low false negatives) while still maximizing trie reduction relative to GCR. We sweep $\tau \in [0.1, 0.6]$ with step size 0.05 and select the highest value that satisfies FNR < 0.05. Thresholds are tuned independently for v1 and v2, since v2 uses step-wise context updates and therefore follows a different pruning dynamic.

3.5.3 Encoder Caching

To reduce repeated computation during beam search, path embeddings $E(\text{str}(p))$ are cached. In v1, caching is done up front because the trie is fixed after construction. In v2, embeddings are cached lazily as new branches are expanded. In both cases, caching avoids redundant encoder calls for previously seen paths. By contrast, the query-context embedding $E(\text{query}(q, y_{<t}))$ must be recomputed whenever the generation prefix changes.

3.6 DCA-Trie v1: Static Semantic Filtering

3.6.1 Design Rationale

DCA-Trie v1 addresses the permissiveness problem while keeping the efficient static-trie design used in GCR. The key idea is to apply semantic filtering once, before decoding begins, using only the input question q. This is useful because many paths are structurally reachable but semantically unrelated to the question. Scoring paths against q during preprocessing removes weak candidates early and reduces the search space before autoregressive generation starts.

Methodologically, this changes the oracle from $f(\mathcal{G}, \mathcal{E}_q)$ to $f(\mathcal{G}, \mathcal{E}_q, q)$ without changing generation-time lookup mechanics. The decoder still queries a precompiled trie, so local lookup remains $O(d)$ as in GCR. Figure 3.3 summarises this offline filtering workflow.

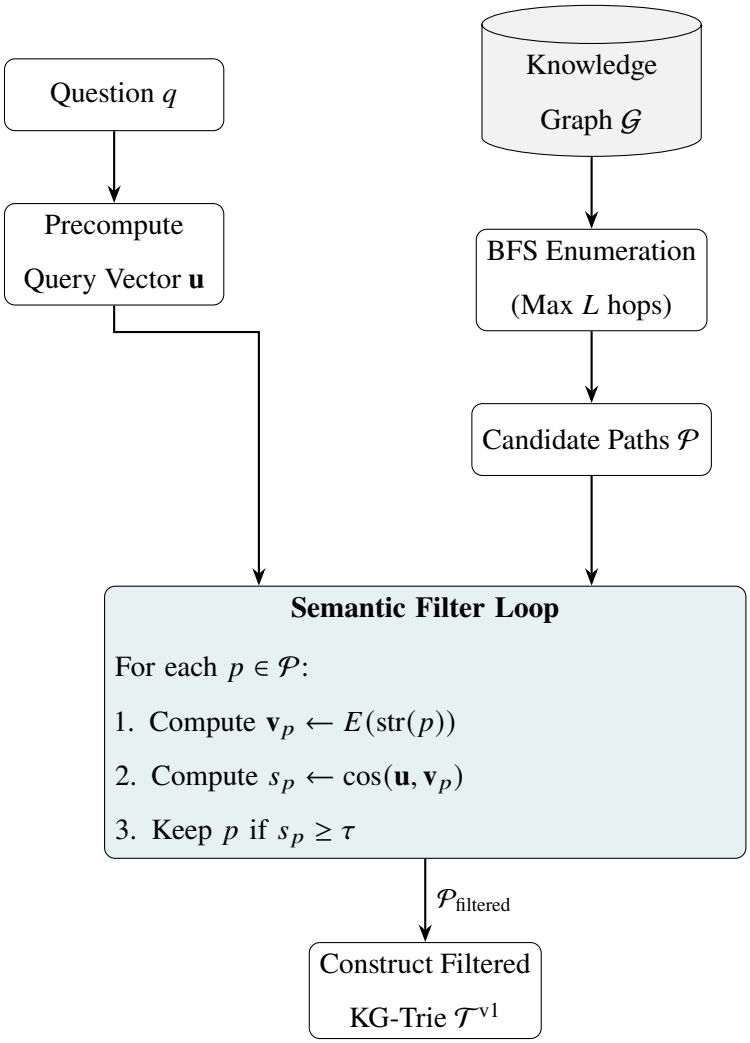

Figure 3.3 Static Semantic Filtering Pipeline for DCA-Trie v1.

3.6.2 Algorithm

Algorithm 1 DCA-Trie v1: Static Semantic Filtering at Construction Time

Require: Knowledge graph $\mathcal{G}$; question entities $\mathcal{E}_q$; question q; max hop depth L; admission threshold τ; sentence encoder E

Ensure: Semantically filtered KG-Trie $\mathcal{T}^{\text{v1}}$

```
Precompute query embedding: u ← E(query(q, ε))      ▷ y_{<t} = ε at construction time
Enumerate paths: P ← BFS(G, E_q, L)                 ▷ All paths within L hops, as in GCR
Score and filter:
for each path p ∈ P do
    v_p ← E(str(p))                                 ▷ Precompute and cache path embedding
    s_p ← cos(u, v_p)
    if s_p ≥ τ then
        Add p to P_filtered
    end if
end for
Construct filtered trie: T^v1 ← BuildTrie(P_filtered)
return T^v1

At generation time (identical to GCR):
for each step t do
    V_t ← TrieLookup(T^v1, y_{<t})
    m_t[v] ← 1[v ∈ V_t] for all v
    α̃_t ← m_t ⊙ LLMLogits(y_{<t}, x)
    y_t ← beam-search-step(α̃_t)
end for
```

Figure 3.4 DCA-Trie v1 algorithm.

3.6.3 Algorithm Description

Algorithm 1 operates in two distinct phases: an offline filtering phase and an online generation phase. To clarify how the mechanism works in practice, the process unfolds in the following

sequence:

1. **Precompute the Query State:** The system first embeds the raw question into a dense vector space. At this construction stage, the generation prefix is empty.

2. **Enumerate the Structural Space:** Using standard breadth-first search (BFS), the system maps out every possible graph path originating from the question entities up to a maximum depth L. This mirrors the baseline GCR approach.

3. **Score and Prune:** Here, the semantic intervention occurs. The system measures the cosine similarity between the question embedding and every enumerated path. Any path scoring below the threshold τ is discarded as contextually irrelevant.

4. **Construct the Static Trie:** The surviving paths are compiled into a static prefix tree ($\mathcal{T}^{\text{v1}}$). This tree maps out the complete, semantically pruned valid search space.

5. **Autoregressive Decoding (Online):** During generation, the LLM queries this pre-built trie at every token step. The trie returns a binary mask forcing the model to only output tokens that advance along the pre-approved paths.

What this demonstrates is that semantic selectivity happens exactly once. By pruning the search space entirely before generation begins, the decoding process itself remains fast, stable, and easily comparable to existing baselines.

3.6.4 Complexity Analysis

This section reports an implementation-level complexity analysis of DCA-Trie v1 under the data-structure assumptions used in this project. Building $\mathcal{T}^{\text{v1}}$ requires an offline cost of $O(|\mathcal{P}| \cdot d_E)$, where $|\mathcal{P}|$ is the number of BFS-enumerated paths and $d_E = 384$ is the embedding dimension. This is a one-time preprocessing cost. During decoding, lookup complexity remains $O(d)$, matching GCR, because trie access uses the same constant-time child lookup structure. Memory usage scales as $O(|\mathcal{P}_{\text{filtered}}| \cdot L)$, which is typically much smaller than GCR's $O(|\mathcal{P}| \cdot L)$ when semantic filtering removes irrelevant paths.

3.6.5 Faithfulness Guarantee

DCA-Trie v1 preserves structural faithfulness by construction. Every retained path in $\mathcal{P}_{\text{filtered}}$ comes from $\text{BFS}(\mathcal{G}, \mathcal{E}_q, L)$, so each triplet (e_{i-1}, r_i, e_i) remains a valid graph edge. Semantic filtering only removes paths; it does not introduce new tokens or edges. Therefore, any token admitted through $\mathcal{T}^{\text{v1}}$ remains structurally grounded, satisfying Condition F under correct token-mask construction and tokenizer alignment.

3.7 DCA-Trie v2: Step-Wise Dynamic Expansion

3.7.1 Design Rationale

DCA-Trie v2 implements the full dynamic oracle in Equation (3.2) by conditioning constraints on the evolving prefix $y_{<t}$. Unlike v1, it does not assume that question-level intent alone is sufficient throughout decoding. As generation progresses, $y_{<t}$ reveals the path decisions already made, so the relevant neighbourhood can change step by step. For example, after selecting *Elvis_Presley* → *starred_in* → *Blue_Hawaii*, plausible next expansions should focus on *Blue_Hawaii* rather than the initial entity set.

Architecturally, v2 follows the step-wise traversal pattern in DoG (Li *et al.*, 2025) but adds semantic gating to each local expansion. After each committed entity e_t, the trie expands only to structurally valid neighbours that also satisfy the threshold τ:

$$\mathcal{T}_{t+1} = \mathcal{T}_t \cup \left\{(e_t,\, r,\, e') : (e_t, r, e') \in \mathcal{T}_{\mathcal{G}} \;\wedge\; \text{score}(e', q, y_{<t}) \geq \tau\right\} \tag{3.11}$$

where $\mathcal{T}_{\mathcal{G}}$ is the set of graph triplets. The core difference from permissive dynamic expansion is therefore not when expansion happens, but how candidates are admitted: every step remains graph-valid and semantically filtered. The full workflow is shown in Figure 3.5.

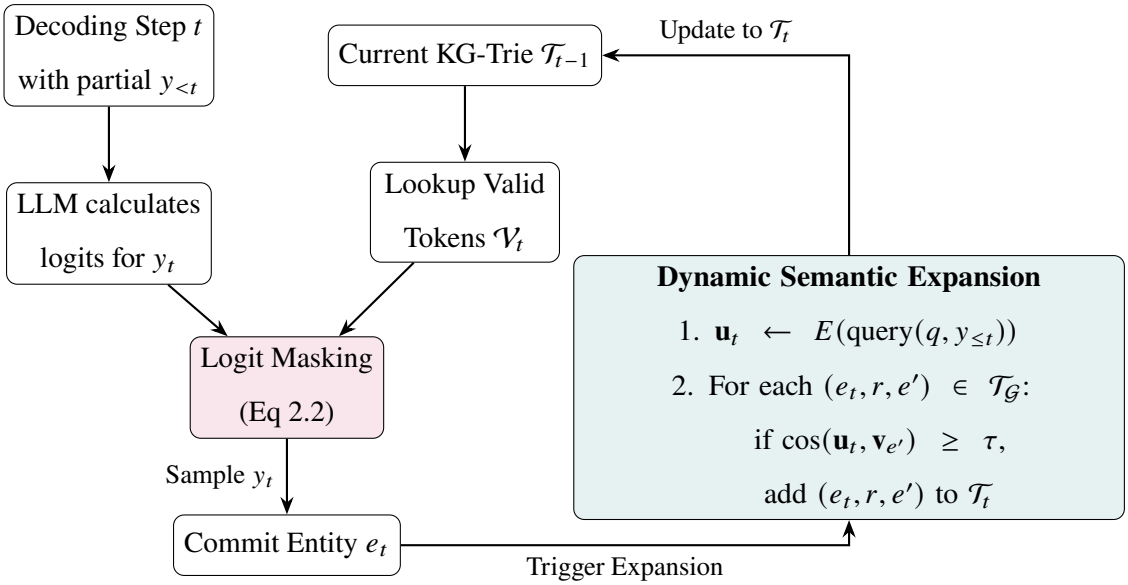

Figure 3.5 Step-Wise Dynamic Expansion Workflow for DCA-Trie v2.

3.7.2 Algorithm

Algorithm 2 DCA-Trie v2: Step-Wise Semantic Expansion

Require: Knowledge graph $\mathcal{G}$; question entities $\mathcal{E}_q$; question q; admission threshold τ; sentence encoder E; LLM with vocabulary $\mathcal{V}$

Ensure: Generated reasoning chain $y_1 \dots y_T$

Initialize trie with entity nodes for $\mathcal{E}_q$: $\mathcal{T}_0 \leftarrow \{e_q : e_q \in \mathcal{E}_q\}$
Precompute query embedding: $\mathbf{u}_0 \leftarrow E(\text{query}(q, \varepsilon))$
for each step $t = 1$ **to** T **do**
    $\mathcal{V}_t \leftarrow \textsc{TrieLookup}(\mathcal{T}_{t-1}, y_{<t})$ ▹ Valid tokens from current trie
    $\mathbf{m}_t[v] \leftarrow \mathbf{1}[v \in \mathcal{V}_t]$ for all v
    $\tilde{\boldsymbol{\alpha}}_t \leftarrow \mathbf{m}_t \odot \text{LLMLogits}(y_{<t}, x)$
    $y_t \leftarrow$ beam-search-step$(\tilde{\boldsymbol{\alpha}}_t)$
    if y_t is an *entity* token e_t **then** ▹ Expansion triggered at entity commits
        Update query embedding: $\mathbf{u}_t \leftarrow E(\text{query}(q, y_{\leq t}))$
        Expand trie:
        for each $(e_t, r, e') \in \mathcal{T}_{\mathcal{G}}$ **do**
            $\mathbf{v}_{e'} \leftarrow E(\text{str}((e_t, r, e')))$ ▹ Cache if previously computed
            $s \leftarrow \cos(\mathbf{u}_t, \mathbf{v}_{e'})$
            if $s \geq \tau$ **then**
                Add (e_t, r, e') to $\mathcal{T}_t$
            end if
        end for
    else
        $\mathcal{T}_t \leftarrow \mathcal{T}_{t-1}$ ▹ No expansion at relation token steps
    end if
end for
return $y_1 \dots y_T$

Figure 3.6 DCA-Trie v2 algorithm.

3.7.3 Algorithm Description

Unlike the static approach in v1, Algorithm 2 interleaves generation and semantic filtering. To navigate how the search space dynamically evolves, the process steps through the following sequence:

1. **Initialize a Minimal State:** The system begins with a bare trie containing only the root question entities, alongside a baseline query embedding.

2. **Generate under Current Constraints:** At any step t, the LLM generates a new token strictly restricted by the current physical boundaries of the trie.

3. **Check for Entity Commitment:** The algorithm branches depending on what the LLM just generated. If it output an intermediate relation token, the trie remains unchanged and decoding continues. However, if the LLM finishes generating an entity token, an expansion phase triggers.

4. **Update the Context:** When an entity is committed, the system recalculates the query embedding. This new embedding fuses the original question with the exact reasoning path the LLM just generated, making the context highly specific to the current state.

5. **Dynamically Expand:** The system retrieves all structurally valid neighbours of the newly committed entity from the master knowledge graph. It scores these candidates against the updated context embedding, discarding the irrelevant ones. The survivors are then spliced into the trie to allow valid future generation.

The implication here is a highly adaptive oracle. Instead of forcing the LLM to choose from a massive, pre-calculated net of paths, the system selectively builds the net just one step ahead of the model's current reasoning state, following its trajectory while maintaining graph validity.

3.7.4 Complexity Analysis

This section reports an implementation-level complexity analysis of DCA-Trie v2 under the same assumptions. In v2, expansion cost occurs only when an entity token is committed, with complexity $O(|\mathcal{N}_{e_t}| \cdot d_E)$, where $|\mathcal{N}_{e_t}|$ is the neighbourhood size of e_t. In practice, semantic

filtering reduces this to approximately $O(|\mathcal{N}_{e_t}^{\text{filtered}}| \cdot d_E)$ with $|\mathcal{N}_{e_t}^{\text{filtered}}| \ll d$ for most nodes. Relation-token steps do not trigger expansion, so their cost remains comparable to standard constrained decoding. Caching further reduces overhead by avoiding repeated encoding of previously seen candidates.

3.7.5 Faithfulness Guarantee

As with v1, v2 preserves structural faithfulness by construction. Every expansion candidate must satisfy $(e_t, r, e') \in \mathcal{T}_{\mathcal{G}}$ in Equation (3.11). Semantic scoring only prunes this structurally valid set; it does not add new graph transitions. Therefore, all admitted tokens remain graph-grounded and Condition F is preserved under correct token-mask construction and tokenizer alignment.

3.8 Baseline Systems

Four system configurations are evaluated to assess the proposed methodology:

CoT Baseline: The same Llama-3.1-8B model (Dubey *et al.*, 2024) used in GCR, run without any KG constraint oracle and prompted with standard chain-of-thought reasoning. This provides the unconstrained reference point.

GCR: The original Graph-Constrained Reasoning framework (Luo *et al.*, 2025a) with static KG-Trie constraints $\mathcal{W}_{\text{val}}(t) = f(\mathcal{G}, \mathcal{E}_q)$. This is the primary constrained baseline. Reproduction targets WebQSP Hits@1 within 1–2% of the reported 92.6%.

DCA-Trie v1: The static semantic filtering variant described in Section 3.6, implemented as a preprocessing step in the GCR pipeline with validation-calibrated threshold τ.

DCA-Trie v2: The dynamic step-wise expansion variant described in Section 3.7, integrated into the decoding process.

To ensure fair comparison, all settings use the same Llama-3.1-8B backbone, the same entity-linking pipeline for $\mathcal{E}_q$, and matched decoding settings (beam width $k = 5$ for constrained systems). This isolates differences to the constraint-oracle design rather than model or preprocessing variation.

3.9 Evaluation Protocol

3.9.1 Datasets

Evaluation is conducted on two standard Freebase-based KGQA datasets (Bollacker *et al.*, 2008):

WebQSP (Yih *et al.*, 2016): 4,737 questions, mostly requiring 1–2 hops. The standard test set is used for evaluation. A separate disjoint set of 100 questions is used only for threshold calibration.

ComplexWebQuestions (CWQ) (Talmor and Berant, 2018): 34,689 questions with higher compositional complexity and hop depths up to 4. This dataset is used in full for evaluation with no overlap with the calibration split.

3.9.2 Evaluation Metrics

Five metrics are used across both datasets:

1. **Hits@1:** The proportion of examples where the top predicted answer matches the gold entity.

2. **F1:** Entity-level overlap score for settings with multiple acceptable answers, following common KGQA evaluation practice (Lan *et al.*, 2022a).

3. **Structural Faithfulness Rate:** The fraction of generated paths whose triplets are all valid in the underlying KG.

4. **Semantic Irrelevance Ratio (SIR):** Computed using Equations (3.6)–(3.7) with fixed reference threshold $\tau_{\text{ref}} = 0.3$.

5. **Average Trie Size Per Step:** The mean size of the valid token set $|\mathcal{V}_t|$ across decoding steps, used as a direct efficiency indicator of constraint selectivity.

3.9.3 Hop-Depth Stratification

All major metrics (except binary structural faithfulness) are also reported by hop depth (1, 2, 3, and 4), consistent with prior KGQA analysis (Lan *et al.*, 2022a). This is important because

permissiveness effects usually increase with hop depth; aggregate-only reporting can hide this behavior.

3.9.4 Experimental Configuration

All experiments are run on a single NVIDIA A100 (40GB). Llama-3.1-8B inference uses 16-bit precision through HuggingFace Transformers (Wolf *et al.*, 2020). Constrained systems use beam width $k = 5$, while the unconstrained CoT baseline uses $k = 1$ to match the original GCR setup. The `all-MiniLM-L6-v2` encoder is served through 'sentence-transformers' (Reimers and Gurevych, 2019) on CPU to avoid GPU contention with LLM decoding. Maximum generation length is set to $3L + 2$, with $L = 2$ for WebQSP and $L = 4$ for CWQ.

3.10 Scope Boundaries and Anticipated Limitations

This study has clear scope boundaries that frame interpretation of the Chapter 4 results:

Single KG and domain: Experiments are limited to Freebase-based KGQA. Generalization to other KGs (e.g., Wikidata and ConceptNet) remains future work.

No model fine-tuning: The Llama-3.1-8B backbone is not fine-tuned. Observed improvements therefore reflect changes in constraint-oracle design rather than parameter adaptation.

Empirical faithfulness evidence: Condition F is assessed empirically by checking generated paths against KG triplets, following established practice in prior work (Luo *et al.*, 2025a; Li *et al.*, 2025). This evidence is interpreted under the same implementation assumptions on valid-token masking and tokenizer alignment.

Threshold as a hyperparameter: The selected τ is tuned on a validation subset and may not transfer optimally across datasets.

False negative risk: Even under Condition P, semantic filtering can remove valid paths, particularly when lexical form and graph semantics diverge. This trade-off is analysed explicitly in Chapter 4.

Latency: Semantic scoring introduces additional compute overhead relative to GCR. Wall-clock latency is therefore reported, but production deployment optimization is outside the scope of this thesis.

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